

[Full thesis title]

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A thesis submitted in partial fulfilment of the requirements for the degree of

Doctor of Philosophy

The University of Sheffield

[Faculty name]

[School name]

[Month and year of submission]

Abstract

Speech models such as text-to-speech and automatic speech recognition typically operate on discretised representations of both acoustic time and computational depth. Although speech is a continuous physical signal, it is usually represented as fixed-rate frames or tokens and processed by a fixed sequence of neural-network layers. In conditional generative speech models, text is also commonly introduced through auxiliary conditioning channels while the source distribution remains an abstract Gaussian, making the interpretation of the resulting conditional likelihood unclear. Continuous-depth models have nevertheless shown that neural transformations can be represented as invertible dynamical systems and evaluated using the change-of-variables formula.

Here, we investigate speech modelling in which both acoustic time and computational depth are continuous. This formulation allows signals to be evaluated at arbitrary temporal resolutions, enabling variable-length utterances to be packed without padding and permitting resolution to increase progressively over depth from sparse linguistic inputs to dense acoustic outputs. More importantly, it creates a route towards defining an invertible transformation from a continuous representation of text to a continuous representation of speech, so that likelihood describes transport between the conditioning and target distributions more directly. Together, continuous time and continuous depth provide a unified framework for adaptive temporal resolution, coarse-to-fine alignment, efficient computation, and probabilistically grounded mappings between language and speech.

Declaration

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Publications arising from the thesis

[Provide details of publications arising from this thesis, or confirm that there are none.]

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Notation

This section provides a concise reference describing notation used throughout this document. If you are unfamiliar with any of the corresponding mathematical concepts, Goodfellow *et al.* (2016) describe most of these ideas in chapters 2–4.

Numbers and Arrays

a	A scalar (integer or real)
$\boldsymbol{a}$	A vector
$\boldsymbol{A}$	A matrix
A	A tensor
$\boldsymbol{I}_n$	Identity matrix with n rows and n columns
$\boldsymbol{I}$	Identity matrix with dimensionality implied by context
$\boldsymbol{e}^{(i)}$	Standard basis vector $[0, \dots, 0, 1, 0, \dots, 0]$ with a 1 at position i
$\text{diag}(\boldsymbol{a})$	A square, diagonal matrix with diagonal entries given by $\boldsymbol{a}$
a	A scalar random variable
$\mathbf{a}$	A vector-valued random variable
$\mathbf{A}$	A matrix-valued random variable

Sets and Graphs

$\mathbb{A}$	A set
$\mathbb{R}$	The set of real numbers
$\{0, 1\}$	The set containing 0 and 1
$\{0, 1, \dots, n\}$	The set of all integers between 0 and n
$[a, b]$	The real interval including a and b
$(a, b]$	The real interval excluding a but including b
$\mathbb{A} \backslash \mathbb{B}$	Set subtraction, i.e., the set containing the elements of $\mathbb{A}$ that are not in $\mathbb{B}$
$\mathcal{G}$	A graph
$Pa_{\mathcal{G}}(\mathrm{x}_i)$	The parents of x_i in $\mathcal{G}$

Indexing

a_i	Element i of vector $\boldsymbol{a}$, with indexing starting at 1
a_{-i}	All elements of vector $\boldsymbol{a}$ except for element i
$A_{i,j}$	Element i, j of matrix $\boldsymbol{A}$
$\boldsymbol{A}_{i,:}$	Row i of matrix $\boldsymbol{A}$
$\boldsymbol{A}_{:,i}$	Column i of matrix $\boldsymbol{A}$
$\mathsf{A}_{i,j,k}$	Element (i, j, k) of a 3-D tensor A
$\mathsf{A}_{:,:,i}$	2-D slice of a 3-D tensor
a_i	Element i of the random vector $\mathbf{a}$

Linear Algebra Operations

$\boldsymbol{A}^\top$	Transpose of matrix $\boldsymbol{A}$
$\boldsymbol{A}^+$	Moore-Penrose pseudoinverse of $\boldsymbol{A}$
$\boldsymbol{A} \odot \boldsymbol{B}$	Element-wise (Hadamard) product of $\boldsymbol{A}$ and $\boldsymbol{B}$
$\det(\boldsymbol{A})$	Determinant of $\boldsymbol{A}$

Calculus

$\frac{dy}{dx}$	Derivative of y with respect to x
$\frac{\partial y}{\partial x}$	Partial derivative of y with respect to x
$\nabla_{\boldsymbol{x}} y$	Gradient of y with respect to $\boldsymbol{x}$
$\nabla_{\boldsymbol{X}} y$	Matrix derivatives of y with respect to $\boldsymbol{X}$
$\nabla_{\mathsf{X}} y$	Tensor containing derivatives of y with respect to X
$\frac{\partial f}{\partial \boldsymbol{x}}$	Jacobian matrix $\boldsymbol{J} \in \mathbb{R}^{m \times n}$ of $f : \mathbb{R}^n \to \mathbb{R}^m$
$\nabla_{\boldsymbol{x}}^2 f(\boldsymbol{x})$ or $\boldsymbol{H}(f)(\boldsymbol{x})$	The Hessian matrix of f at input point $\boldsymbol{x}$
$\int f(\boldsymbol{x}) d\boldsymbol{x}$	Definite integral over the entire domain of $\boldsymbol{x}$
$\int_{\mathbb{S}} f(\boldsymbol{x}) d\boldsymbol{x}$	Definite integral with respect to $\boldsymbol{x}$ over the set $\mathbb{S}$

Probability and Information Theory

$\mathrm{a} \perp \mathrm{b}$	The random variables a and b are independent
$\mathrm{a} \perp \mathrm{b} \mid \mathrm{c}$	They are conditionally independent given c
$P(\mathrm{a})$	A probability distribution over a discrete variable
$p(\mathrm{a})$	A probability distribution over a continuous variable, or over a variable whose type has not been specified
$\mathrm{a} \sim P$	Random variable a has distribution P
$\mathbb{E}_{\mathrm{x} \sim P}[f(x)]$ or $\mathbb{E} f(x)$	Expectation of $f(x)$ with respect to $P(\mathrm{x})$
$\mathrm{Var}(f(x))$	Variance of $f(x)$ under $P(\mathrm{x})$
$\mathrm{Cov}(f(x), g(x))$	Covariance of $f(x)$ and $g(x)$ under $P(\mathrm{x})$
$H(\mathrm{x})$	Shannon entropy of the random variable x
$D_{\mathrm{KL}}(P \| Q)$	Kullback-Leibler divergence of P and Q
$\mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution over $\boldsymbol{x}$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$

Functions

$f : \mathbb{A} \to \mathbb{B}$	The function f with domain $\mathbb{A}$ and range $\mathbb{B}$. The type (scalar/vector/matrix) of range is indicated by f
$f \circ g$	Composition of the functions f and g
$f(\boldsymbol{x}; \boldsymbol{\theta})$	A function of $\boldsymbol{x}$ parametrized by $\boldsymbol{\theta}$. (Sometimes we write $f_\theta(\boldsymbol{x})$ or omit the argument $\boldsymbol{\theta}$ to lighten notation)
$\log x$	Natural logarithm of x
$\sigma(x)$	Logistic sigmoid, $\dfrac{1}{1+\exp(-x)}$
$\text{softplus}(x)$	Softplus, $\log(1+\exp(x))$
$\|\|\boldsymbol{x}\|\|_p$	L^p norm of $\boldsymbol{x}$
$\|\|\boldsymbol{x}\|\|$	L^2 norm of $\boldsymbol{x}$
x^+	Positive part of x, i.e., $\max(0, x)$
$\mathbf{1}_{\text{condition}}$	is 1 if the condition is true, 0 otherwise

Sometimes we use a function f whose argument is a scalar but apply it to a vector, matrix, or tensor: $f(\boldsymbol{x})$, $f(\boldsymbol{X})$, or $f(\mathbf{X})$. This denotes the application of f to the array element-wise. For example, if $\mathbf{C} = \sigma(\mathbf{X})$, then $C_{i,j,k} = \sigma(X_{i,j,k})$ for all valid values of i, j and k.

Datasets and Distributions

p_{data}	The data generating distribution
$\hat{p}_{\text{data}}$	The empirical distribution defined by the training set
$\mathbb{X}$	A set of training examples
$\boldsymbol{x}^{(i)}$	The i-th example (input) from a dataset
$y^{(i)}$ or $\boldsymbol{y}^{(i)}$	The target associated with $\boldsymbol{x}^{(i)}$ for supervised learning
$\boldsymbol{X}$	The $m \times n$ matrix with input example $\boldsymbol{x}^{(i)}$ in row $\boldsymbol{X}_{i,:}$

My custom notation

$\mathbf{S}_{\boldsymbol{\theta}}$	Score model
$\mathbf{D}_{\boldsymbol{\theta}}$	Data prediction model
$\mathbf{F}_{\boldsymbol{\theta}}$	Flow-matching model (vector field estimator)

Chapter 1

Introduction

1.1 Background

A central pattern in the advancement of deep learning is crafting methods and architectures for the task and data at hand. An image is structured in two dimensions, thus a two-dimensional convolutional kernel is a sensible modelling choice. Time series such as speech waveforms often have a causal relationship between past and future events, and recurrent neural networks share this property. Transformers build on this with parallel training computation over the sequence dimension and longer context modelling. These choices carefully consider the data, the form of the task, and scale required.

Typically, these models are constrained to discrete sequences of hidden layers. This constraint lacks support for modelling continuous dynamics, which may be more generalisable since a continuous process can be identical to a discrete process, but the reverse is not true. Recurrent nets may be applied to continuous time series, but they cannot model a continuous process. Overcoming these limitations has become a topic to be investigated.

Recent years have witnessed a growing academic interest in *neural differential equations*. These models parametrise a vector-field with a neural network, enabling a continuous hidden state. The result has two key benefits for information processing: 1) **continuous-depth** allows scalable and adaptive models, and 2) alternatively the continuous dynamics can be tied to the continuity of the data: these **continuous-time** models can process streams of data that arrive at irregular times. The generalisations of these two claims to applied tasks are the focus of this thesis.

Pre-requisites This thesis assumes familiarity with deep learning, differential equations, and speech technology.

Figure 1.1: Caption

1.2 A brief review of neural differential equations

The development of a deep learning model considers the definition of a successive hidden state

$$\boldsymbol{h}_{u+1} \coloneqq \quad ?. \tag{1.1}$$

A strong option is a residual layer

$$\boldsymbol{h}_{u+1} \coloneqq \boldsymbol{h}_u + \boldsymbol{f}(\boldsymbol{h}_u; \theta_u), \tag{1.2}$$

which is the current hidden state plus the output of a learned neural network $\boldsymbol{f}$, with trainable parameters θ_u. This equation is similar to an Euler discretisation with step-size $\Delta u = 1$

$$\boldsymbol{h}_{u+1} = \boldsymbol{h}_u + \boldsymbol{f}(\boldsymbol{h}_u; \theta_u) \cdot 1. \tag{1.3}$$

Through this observation, $\boldsymbol{f}$ becomes a *vector field*. This vector field is variant to depth u through specific parameters at each layer θ_u, although a typical vector field would simply depend on depth u directly. In addition to depending on u, another relaxation would be to support continuous u and arbitrary step-size Δu,

$$\boldsymbol{h}_{u+\Delta u} \coloneqq \quad ?. \tag{1.4}$$

The desired relaxations can be supported by a continuous Euler update

$$\boldsymbol{h}_{u+\Delta u} \coloneqq \boldsymbol{h}_u + \boldsymbol{f}(\boldsymbol{h}_u, u; \theta) \cdot \Delta u. \tag{1.5}$$

This form of update with a parametrised vector field is known as a *neural ordinary differential equation*. Formally, this leads to the definition of the hidden state as

$$\boldsymbol{h}_u = \boldsymbol{h}_0 + \int_0^u \boldsymbol{f}_\theta(\boldsymbol{h}_s, s)\mathrm{d}s. \tag{1.6}$$

A neural ODE is a flexible architecture that is easier to work with on certain tasks than non-continuous models. For example, a continuous-depth generative model can be constructed by bounding u between 0 and U, then modelling a flow between a simple distribution at $u = U$ and a data distribution at $u = 0$. This flow is optimised end-to-end by allocating a single set of parameters θ to support general u, rather than optimising a subset of the flow at discrete points.

This form of generative model is a *continuous normalising flow*, and can be trained by maximum likelihood or a variational lower bound. There is much discussion on how to optimise a Neural ODE. A usual distinction is simulation vs simulation-free methods. Methods such as the continuous adjoint simulate the ODE forwards and backwards to compute gradients, whereas others construct a reference flow which can be sampled analytically and regressed. This thesis considers the many ways to optimise a simulation-free probabilistic flow e.g. diffusion, flow-matching.

Another perspective considers the flexibility of continuously described dynamics, which is useful for describing continuous dynamics. For example, the hidden state $\boldsymbol{h}_u$ can be evaluated for any u, which allows the generation of irregularly sampled time-series e.g. $u \in \{1, 3, 4, 6, 7, 11\}$. Examples of irregularly sampled data include medical data, stock market trading, and text labels aligned to speech recordings.

1.3 Known vs unknown generalisations

Continuous normalising flows such as diffusion models have become strong models for TTS, but their probabilistic strength is unmeasured. Models such as Grad-TTS have scored higher in subjective evaluation, and they can alternatively be evaluated objectively by likelihood. The likelihood of the data determined by the model determines how important the probabilistic component is within the pipeline. The meaning of "likelihood" in a task such as TTS is naturally the likelihood of an acoustic observation given a text source. Models have not been evaluated this way which leads to a gap in research.

Continuous normalising flows have become strong models for DNS, but whether designs that better fit the nature of the task are better is unknown. Diffusion models form a flow between standard normal source and a target data distribution by simulating reverse Brownian motion. Schrödinger bridges alternatively support a data source, but maintain the assumption that the flow should be Brownian. It is uncertain to what extent Brownian motion is necessary for high-quality DNS.

Continuous-time models have been used for speech tasks with end-point optimisation ($\boldsymbol{h}_U$), but tasks such as TTS and ASR optimise intermediate states $\boldsymbol{h}_u$. An additional complication is the coupling of a continuous speech signal and a discrete sequence of tokens. This outlines a gap in the known generalisations of continuous-time models applied to speech tasks.

1.4 Research questions

1.4.1 Continuous-depth

There is little to no detailed investigation into the generalisations of NDEs on speech processing tasks such as speech synthesis, recognition, and noise suppression. Although breakthrough implementations of CNFs for speech processing exist, there is a lack of studies investigating the likelihood of these models. Furthermore, no previous study has analysed the impact of Brownian motion in noise suppression, a task that modifies the typical Gaussian-to-data formula to data-to-data. This forms two research questions.

RQ1 How well does diffusion model likelihood correspond to expected behaviour in image and speech synthesis?

Diffusion models are an efficient way to train CNFs (equation 1.6), and have been used for image and speech synthesis. How probabilistic is the model in this case? What are the semantics

of "likelihood" in these contexts? Does a picture of a red ball with caption "blue cube" have a low likelihood? Do a spoken utterance and its transcript have a high likelihood?

RQ2 How well do CNFs generalise to data-to-data tasks such as deep noise suppression, and how important is Brownian motion?

DNS deals with transforming a noise-speech mixture into a clean speech signal. Diffusion models exploit Brownian motion to learn this function. How necessary is Brownian motion? Flow matching is an alternative method that supports data-to-data transformations without Brownian motion. How well does this fit the DNS task?

1.4.2 Continuous-time

RQ3 What role can neural CDEs play in speech synthesis?

Is modelling irregularly sampled time-series useful for describing text-speech alignment? Is the propagation of duration information useful? What aspects of the design space of rough path theory affect synthesis quality? Do CDEs allow better synthesis control?

RQ4 To what extent can memory requirements be reduced by using continuous-time backpropagation for non-endpoint optimisation of neural ODEs?

Does a neural ODE require fewer parameters? Is the model quality the same? Does it take longer to train than gradient check-pointing or recurrent modules with backpropagation through time?

1.5 Contributions

Chapter X answers RQ1 with an empirical investigation into how likelihood is computed on different data setups. It is shown that less accurate transcriptions have higher likelihood. For image synthesis, images with correct captions did not have a higher likelihood than those with incorrect captions.

Chapter X answers RQ2 by constructing a flow matching speech denoiser and comparing it with a Schrödinger bridge model. Schrödinger bridges can be thought of as a data-to-data CNF whose intermediate state moves with Brownian motion, whereas flow matching has a linear interpolating state. Experiments show little evidence motivating Brownian motion. Additional inference schemes are tested, where it is suggested that once a flow matching model is trained, it is better to directly predict the data from the model rather than solve an ODE.

Chapter X forms RQ3 through emotional TTS. It is found that human judgment to the intended emotion style is better correlated with a continuous-time mechanism in place.

Chapter X answers RQ4 by training NDE decoders on top of pretrained ASR models, or learning in an end-to-end manner.

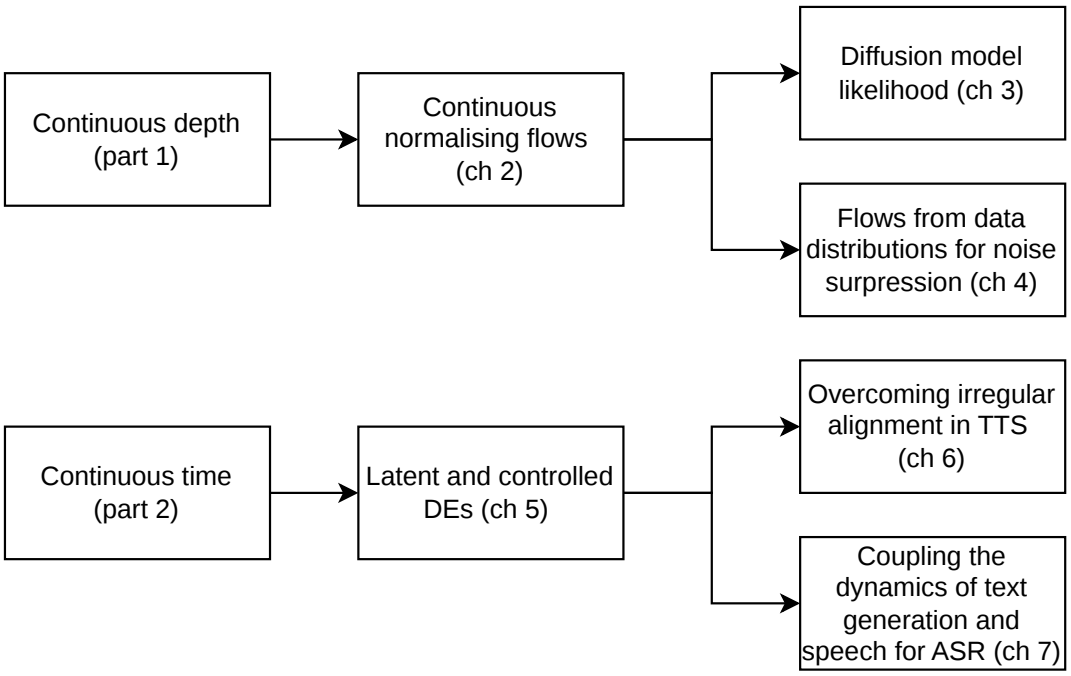


Figure 1.2: Enter Caption

1.6 Dissertation outline

The rest of this dissertation is organised into two parts. The first part is concerned with CNFs and studies continuous-depth models for speech and image data. Chapter X provides a background on training CNFs with diffusion, and how this can be used for speech and image synthesis. Chapter X examines the empirical properties of diffusion likelihood. Chapter X builds further on CNFs by reviewing methods for data-to-data processes with Schrödinger bridge and flow matching models. These models are then used to explore the path design in continuous depth speech denoisers by observing the use of Brownian motion in chapter X. The second part considers continuous-time models for speech synthesis and recognition. Chapter X introduces a background to neural CDEs, which allow a neural ODE to react to an input stream of data. Neural CDEs are then used in chapter X and X for the aforementioned tasks. Concluding remarks on the perspective developed in this thesis are given in chapter X.

1.7 Publications

Chapter 2

Background

2.1 Normalising flows and continuous-depth

A generative model is a sampleable probability distribution $p_0(x)$ of an underlying data distribution p_{data}. Normalising flows are a type of generative model that learn a transform f (flow) from the data to a simple distribution (normal). The inverse of this transform then becomes a generative model. The rest of this section will explain the details of normalising flows and motivation for continuous-depth.

Normalising flows are appropriately named given that they learn a flow (transform) to a normal (simple distribution), and hence "normalise" p_{data}. Key to this definition is the change-of-variables formula (**??**). For a density $p_0(\mathrm{x})$, the change-of-variables formula states that the density of any $\mathrm{z} = \mathrm{f}^{-1}(\mathrm{x})$ is

$$p_0(\mathrm{x}) = p_1(\mathrm{z}) \left| \det \frac{\partial \mathrm{f}(\mathrm{z})}{\partial \mathrm{z}} \right|^{-1} \tag{2.1}$$

where p_0 models p_{data} and p_1 is a simple density, for example a normal density. Since $\mathrm{z} = \mathrm{f}^{-1}(\mathrm{x})$, the equivalent expression in terms of x is

$$p_0(\mathrm{x}) = p_1(\mathrm{f}^{-1}(\mathrm{x})) \left| \det \frac{\partial \mathrm{f}^{-1}(\mathrm{x})}{\partial \mathrm{x}} \right| \tag{2.2}$$

This has the log form

$$\log p_0(\mathrm{x}) = \log p_1(\mathrm{f}^{-1}(\mathrm{x})) + \log \left| \det \frac{\partial \mathrm{f}^{-1}(\mathrm{x})}{\partial \mathrm{x}} \right| \tag{2.3}$$

The density p_0 can be trained by specifying a transform $\mathrm{f}_{\boldsymbol{\theta}}$ and maximising the expected log likelihood over data samples $\mathrm{x} \sim p_{\text{data}}$:

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \mathbb{E}_{\mathrm{x} \sim p_{\text{data}}} \left[\log p_0(\mathrm{x}; \boldsymbol{\theta}) \right] \tag{2.4}$$

The population optimum matches p_0 to p_{data} when the model family contains the data density and the optimisation reaches a global optimum; finite data and model capacity need not yield an exact match.

A large factor in the success of a normalising flow is the choice of transform f. It must be invertible, expressive, and computationally tractable. With a discrete composition such as $f_1 \circ f_2 \circ f_3$, evaluating equation 2.3 requires the log determinant of each layer Jacobian. A dense determinant generally costs cubic time in the state dimension, although specialised architectures can make it cheaper.

A central motivation for continuous-depth normalising flows is that the change-of-variables formula can be expressed using the divergence of a vector field. For the ODE $\frac{\mathrm{dx}_u}{\mathrm{d}u} = \mathrm{f}_{\boldsymbol{\theta}}(\mathrm{x}_u, u)$, the instantaneous-change-of-variables formula Chen *et al.* (2018); Song *et al.* (2021b) gives

$$\log p_0(\mathrm{x}) = \log p_1(\mathrm{x}_1) + \int_0^1 \nabla \cdot \mathrm{f}_{\boldsymbol{\theta}}(\mathrm{x}_u, u)\mathrm{d}u \tag{2.5}$$

The formula assumes a well-posed ODE flow (in particular, conditions ensuring existence and uniqueness). Computation still requires solving the ODE and evaluating its divergence; the cost depends on the solver and the method used to estimate the divergence.

A remaining critique would be that the ODE must be solved between 0 and 1 for every optimisation iteration. In other words, the ODE must be simulated during training. This problem has been alleviated through *simulation-free* training, which aims to sample intermediate states $\mathrm{x_u}$ through closed-form solutions and optimises the vector-field at the sampled point u. A method for defining differential equations from a structured initialisation towards a normal distribution is through stochastic differential equations.

2.2 Stochastic differential equations

2.2.1 The Wiener process

The Wiener process (also called Brownian motion) can be thought of as a random walk continuous in u. It starts from $\mathrm{w}_0 = 0$ and has increments $\mathrm{w}_{u+\Delta u} - \mathrm{w}_u \sim \mathcal{N}(0, \Delta u)$. The inspiration for this process comes from Robert Brown observing the motion of pollen in water. A pollen particle moves randomly in all directions; this is known as diffusion.

The Wiener process is useful here as it can be used to predict the likelihood of a particle position at a given time. The Wiener process is also easy to sample from in closed-form

$$\mathrm{w}_u \sim \mathcal{N}(0, u), \tag{2.6}$$

but only describes evolution of samples from the origin. For more complicated processes, more mechanisms are required.

2.2.2 Itô SDE

An Itô SDE combines an ordinary drift with stochastic diffusion:

$$\mathrm{dx} \coloneqq f(\mathrm{x}, u)\mathrm{d}u + g(\mathrm{x}, u)\mathrm{dw}_u \tag{2.7}$$

These processes are the first example SDEs to be introduced in this dissertation. For example, take an Ornstein-Uhlenbeck process which aims to model a stationary process. The key property is modelling reversion back to the mean despite diffusion perturbing samples randomly

$$\mathrm{dx} \coloneqq \gamma(y - \mathrm{x})\mathrm{d}u + \sigma\mathrm{dw}_u. \tag{2.8}$$

SDEs are characterised by a deterministic component *drift*, and non-deterministic diffusion. This process scales reversion to the selected mean y with γ and diffusion with σ. Although less obvious than the Wiener process, equation 2.8 also has a closed-form solution

$$\mathrm{x}_u \sim \mathcal{N}\left(y + e^{-\gamma u}(\mathrm{x}_0 - y), \frac{\sigma^2}{2\gamma}(1 - e^{-2\gamma u})\right), \qquad \gamma > 0. \tag{2.9}$$

As $u \to \infty$, this conditional distribution converges to the stationary distribution $\mathcal{N}(y, \sigma^2/(2\gamma))$.

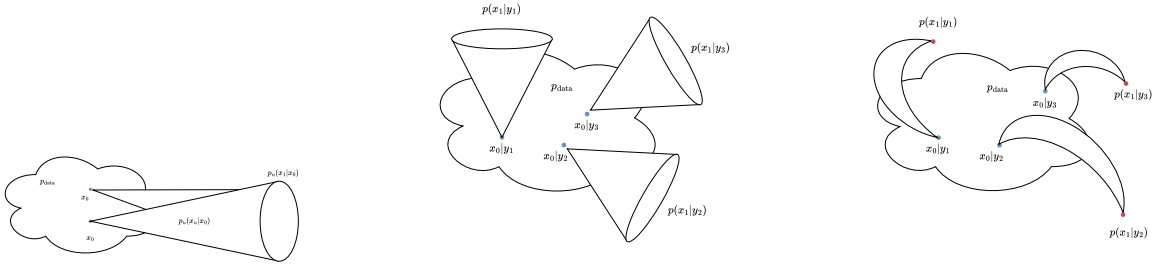
2.3 Deep equilibrium models

This thesis focusses on neural DEs, but the reader should be aware that this is not the only way to achieve "implicit layers". Neural DEs are implicit because the parameters describe a single layer which can be used by black-box equation solvers to simulate depth. Another way to simulate depth is with DEQs.

DEQs are solved by finding an equilibrium point, where the output of the layer is equal to or within a small margin of error from the input.

Part I

Continuous depth



(a) Cartoon 2 (b) Cartoon 3 (c) Cartoon 4

Figure 2.1: Cartoons 2–4.

Forward

Part I explores continuous-depth models, including score-based generative models, Schrödinger bridges, and flow-matching models. Score-based models use SDEs to connect a data distribution to a tractable reference distribution through diffusion. Schrödinger bridges can connect prescribed endpoint distributions; flow-matching methods can be formulated for a range of prescribed probability paths.

Some methods provide *simulation-free* objectives for training continuous normalising flows (CNFs), avoiding a black-box ODE solve at every training step. Likelihood evaluation and sampling can still require numerical ODE integration. The first example considered here is included in chapter 3.

For models that define a normalised density, likelihoods can be evaluated and used to study model behaviour, although likelihood alone does not diagnose generalisation errors. Chapter **??** presents experiments involving likelihood, including whether a model distinguishes an image of a red ball paired with the correct caption from one paired with "blue cube".

Schrödinger bridges (SBs) and conditional flow matching (CFM) are related design choices, not mutually exclusive methods. SBs define a stochastic reference process and seek a path close to it under endpoint constraints; flow matching trains a vector field against a chosen probability path. A path need not be optimal transport unless the construction imposes that property. Stochasticity can act as an entropic regulariser, while excessive input variance may make training harder. The methods are outlined in chapter 3 and evaluated in chapter **??**.

An additional investigation on SBs and CFMs concerns their ability to directly predict data from an input sample. This removes the continuous-depth aspect to these models, and introduces

similarities to curriculum learning and data augmentation. A typical simulation-free objective for SB and CFM involves predicting a data sample given a sample from the probability path. If by this training the end-to-end data prediction is accurate, then the advantage of continuous-depth may not be increased capacity but instead through improved parameter optimisation.

This thesis makes an important distinction of the variable u. In the literature it is common to use t to denote depth as this is consistent with the differential equations literature. The use of t for depth becomes problematic in this thesis as the concept of time is better suited to the time-progression of a series, as is later described in part II. As a result, u is used in part I to represent the level of depth through the DE, and t is used to represent time within a series.

Chapter 3

Simulation-free training of continuous normalising flows

3.1 From simulation-based to simulation-free training

Continuous normalising flows (CNFs) connect a source distribution to a data distribution through a depth-dependent vector field. Training through numerical integration can require repeated forward and backward solves. Simulation-free objectives instead sample intermediate states from a prescribed probability path and regress an available conditional target. Sampling and likelihood evaluation can still require numerical integration.

This chapter organises these methods around probability paths, prediction targets, conditioning, and inference. Throughout, $u = 0$ denotes the data endpoint and $u = 1$ the source endpoint; generation proceeds towards decreasing u.

Figure 3.1 provides a visual overview of the paths considered below: mean-reverting variance-preserving diffusion (MR-VP), variance-exploding Schrödinger bridges (SB-VE), static-variance paths (SB-SV), and the linear interpolation paths used for flow matching (labelled ICFM in the figure).

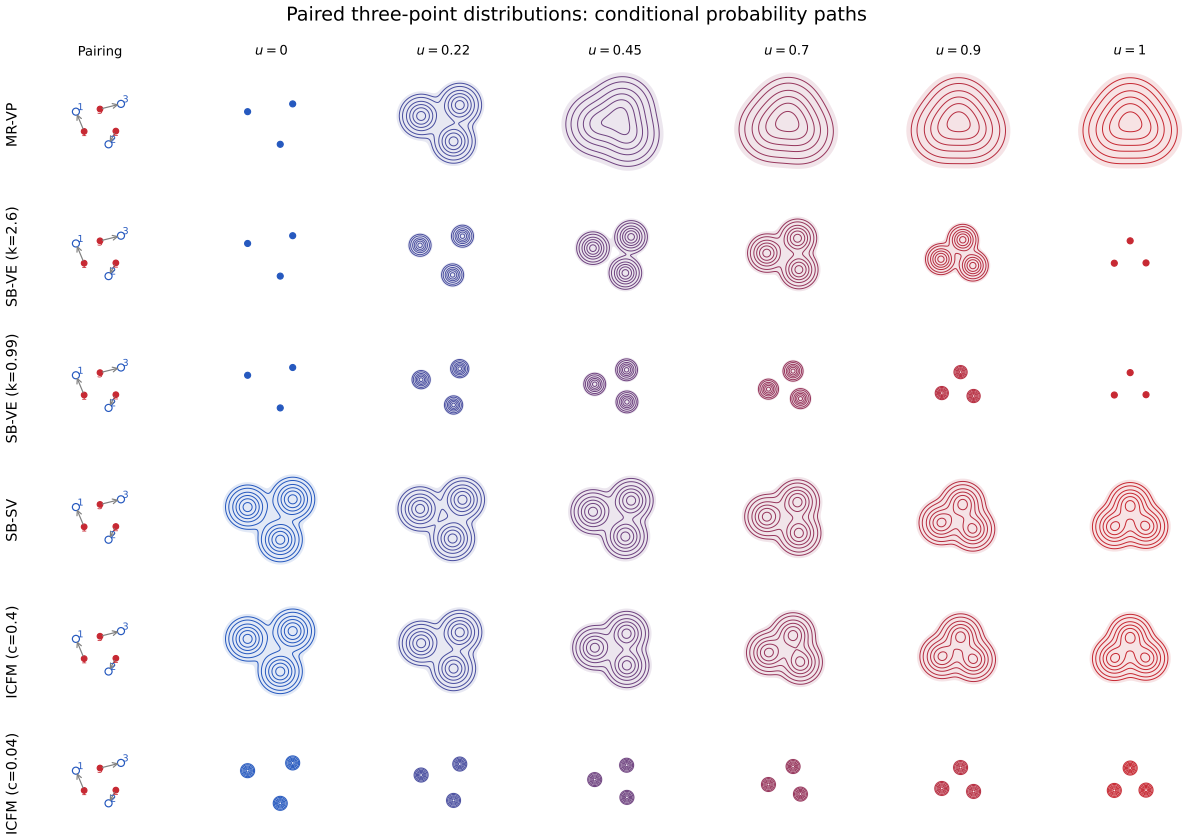


Figure 3.1: Probability-path overview for three paired target and source points. Each row shows density contours at increasing depth u; generation proceeds from right to left. MR-VP approaches a Gaussian mixture about the source points, SB-VE has exact point-mass endpoints, and SB-SV and the flow-matching paths retain Gaussian smoothing at both endpoints. The two SB-VE schedules and two flow-matching variances illustrate their effects on the intermediate densities. Here ICFM denotes the linear, constant-variance path form used with the displayed pairing, rather than independent endpoint sampling.

3.2 Probability paths and endpoint distributions

The central tool for simulation-free training is the definition of a probability path p_u. The conditional paths considered here are Gaussian, with intermediate states $\mathbf{X}_u \sim p_u$ sampled as

$$\mathbf{X}_u = \mathbf{M}_u + \Sigma(u)^{1/2}\mathbf{Z}, \qquad \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{3.1}$$

The mean $\mathbf{M}$ and covariance Σ depend on the chosen construction and its conditioning variables. A typical mean function is an interpolation of endpoint samples such as

$$\mathbf{M}(u, \mathbf{X}, \mathbf{Y}) \coloneqq a(u)\mathbf{X} + b(u)\mathbf{Y} \tag{3.2}$$

where $a(0) = b(1) = 1$ and $a(1) = b(0) = 0$. When $\mathbf{Y} = 0$, the marginal at $u = 1$ becomes similar to a standard normal $p_1(\mathbf{X}_1) \simeq \mathcal{N}(\mathbf{0}, \Sigma)$. The boundaries p_0 and p_1 are known as the *target* and *source* distribution respectively. A path derived from a data endpoint is known as a *conditional* path $p_u(\mathbf{X}_u|\mathbf{X}_0)$, which may also be conditioned on $\mathbf{X}_1$ as well.

For paired endpoints, their joint law is denoted by $\pi(\mathbf{X}, \mathbf{Y})$. This contrasts the original idea of a normalising flow, as now the flow is a coupling between two potentially non-normal distributions. Despite this, such flows still lie within the CNF framework.

Conditional paths are used as a reference for learning a marginal path. This aspect is what makes training simulation-free, as the conditional path can be sampled at any u during training, and the marginal path is optimised for any u. This format is useful, and is discussed in section **??**.

A contrast simulation-free training brings is through the optimisation objective. When simulating the CNF, likelihood is maximised directly, and the learnt path is a result of the optimisation algorithm. A probability path instead acts as a target in which a sufficiently trained estimator can retrieve the data marginal. Yet, this is achieved by learning the probability path rather than likelihood maximisation. A useful feature of simulation-free training is that the likelihood density is still tractable through the ICOV formula.

These paths typically take the form of a closed expression of a DE, as this guarantees that there is a valid trainable vector field. Another common factor is the use of Gaussian probability paths. Although this may seem as an incorrect assumption of the underlying data, by considering the data as a mixture of Gaussians centred at the data points, the smaller the variance of each Gaussian, the closer to the data distribution the path endpoint becomes.

Score-based diffusion models were first developed as a method for training a standard CNF, but many proposed modifications have been made to support extended tasks and data such as conditional generation and paired endpoints. On the other hand, Schrödinger bridges specify a path problem with endpoint constraints and reference dynamics e.g. go from point A to point B and follow a diffusion-like path. Finally, flow matching also specify a path between endpoints, but no restrictions on the dynamics, thus allowing straight-line paths, which are optimal in the sense that they require the least amount of movement.

The next sections detail the mechanics of these probability paths and how to construct them. The order these are presented in follows the chronology of "breakthrough" findings. First diffusion

models proved that SDEs can be used simulation-free training of CNFs with a tractable LL, then SBs use forward-backward SDEs for exact-endpoint interpolation, finally flow-matching shows a generalised view through ODEs. A key observation made in this thesis is the amount of stochasticity each method has, and its impact on training and sampling.

Mention something about LL here?

3.2.1 Optimal transport

I feel this discussion is important, yet I'm unsure where to put it, maybe summary at the end?... From an OT POV, FM can be seen as a solution to the dynamic OT problem, and SB is an entropic regularisation of this. When dealing with paired endpoints, the coupling is already resolved. With this, only the minimum kinetic energy component of the dynamic OT problem is optimisable, of which FM fully addresses but DM and SB address with added noise. The benefit of added noise is an open topic.

$$p_0(\mathbf{X}_0|\mathbf{Y}) = \iint p_0(\mathbf{X}_0|\mathbf{X}_1)\mathcal{N}(\mathrm{d}\mathbf{X}_1|\mathbf{Y}, \Sigma_1)\mathrm{d}p_{\text{data}}(\mathbf{X}_0, \mathbf{Y}) \tag{3.3}$$

$$p(x0|x1)p(x1|y) \tag{3.4}$$

3.2.2 Diffusion paths

Recall that the goal of a CNF is to learn a probability path between a source distribution and target distribution. This is formalised as a time varying probability path with boundaries at the desired distributions e.g. $\mathcal{N}(\mathbf{0}, \Sigma) \to p_{\text{data}}$. As indicated earlier, the reverse of this path can be described as a diffusion process. Starting from a structured initialisation at $u = 0$, the Wiener process $\mathbf{W}$ shifts points over time, and at the terminal time $u = 1$ the distribution has a similar density to a normal distribution. The formal description of this begins with an Itô SDE

$$\mathrm{d}\mathbf{X} \coloneqq \mathbf{F}(\mathbf{X})\mathrm{d}u + g(u)\mathrm{d}\mathbf{W}_u \tag{3.5}$$

where an initial value at $u = 0$ is sampled from the data $\mathbf{X}_0 \sim p_{\text{data}}(\mathbf{X}_0)$ and at $u = 1$ the probability path p_u becomes similar to a normal. Any intermediate and final samples can be computed with the following solution

$$\mathbf{X}_u = \mathbf{X}_0 + \int_0^u \mathbf{F}(\mathbf{X}_s, s)\mathrm{d}s + \int_0^u g(s)\mathrm{d}\mathbf{W}_s, \tag{3.6}$$

either with black-box SDE solvers or in closed form.

The reverse process is core to finalising the CNF construction, which is written by the result of Anderson (1982)

$$\mathrm{d}\mathbf{X} = \left[\mathbf{F}(\mathbf{X}) - g(u)^2\nabla \log p_u(\mathbf{X})\right]\mathrm{d}u + g(u)\mathrm{d}\bar{\mathbf{W}}_u \tag{3.7}$$

where $\bar{\mathbf{W}}_u$ is the reverse Wiener process. The gradient of the log-density $\nabla \log p_u(\mathbf{X}_u)$ is known

as the score, and models that estimate it are known as score-based generative models.

Figure to N(0,1).

3.2.3 Mean reversion and variance preservation

Define a function of u, known as a β-schedule

$$\beta(u) \coloneqq \beta_{\min} + \big(\beta_{\max} - \beta_{\min}\big)u \tag{3.8}$$

This schedule is used to construct a variance preserving SDE

$$\mathrm{d}\mathbf{X} \coloneqq \frac{1}{2}(\boldsymbol{Y} - \mathbf{X}_u)\beta(u)\mathrm{d}u + \sqrt{\beta(u)}\mathrm{d}\mathbf{W}_u. \tag{3.9}$$

A consequence of the β-schedule is that the strength of the mean-reversion towards $\boldsymbol{Y}$ increases linearly with u. Secondly, the rate at which diffusion contributes variance increases linearly with u. Together, the drift and diffusion terms are balanced by the coefficients $\frac{1}{2}\beta(u)$ and $\sqrt{\beta(u)}$, resulting in a terminal distribution p_1 with approximately unit covariance given $\boldsymbol{Y}$.

Substituting drift and diffusion into equation 3.7, the reverse SDE is

$$\mathrm{d}\mathbf{X} = \left(\frac{1}{2}(\boldsymbol{Y} - \mathbf{X}_u) - \nabla \log p_u(\mathbf{X}_u)\right)\beta(u)\mathrm{d}u + \sqrt{\beta(u)}\mathrm{d}\hat{\mathbf{W}}_u \tag{3.10}$$

A useful property of this SDE is that it's closed-form solution is known for an initial $\mathbf{X}_0$.

$$p_u(\mathbf{X}_u|\mathbf{X}_0) = \mathcal{N}(\mathbf{X}_u; \mathbf{M}(\mathbf{X}_0, u, \boldsymbol{Y}), \Sigma(u)) \tag{3.11}$$

$$\mathbf{M}(\mathbf{X}_0, u, \boldsymbol{Y}) = \left(1 - e^{-\frac{1}{2}\int_0^u \beta(s)\mathrm{d}s}\right)\boldsymbol{Y} + e^{-\frac{1}{2}\int_0^u \beta(s)\mathrm{d}s}\mathbf{X}_0 \tag{3.12}$$

$$\Sigma(u) = \left(1 - e^{-\int_0^u \beta(s)\mathrm{d}s}\right)\mathbf{I} \tag{3.13}$$

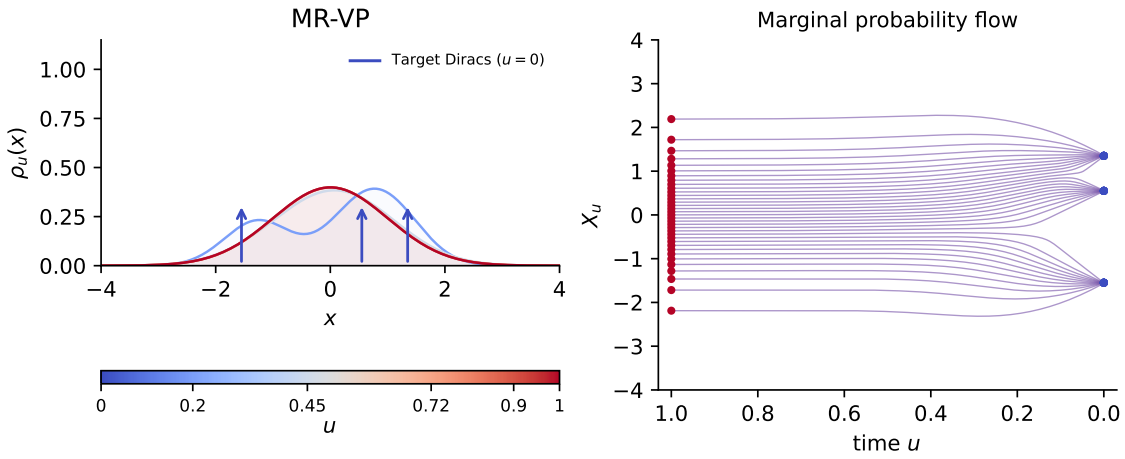
$$\int_0^u \beta(s)\mathrm{d}s = \beta_{\min}u + \frac{1}{2}(\beta_{\max} - \beta_{\min})u^2 \tag{3.14}$$

This allows sampling from p_u instead of solving equation 3.6, resulting in *simulation-free* training.

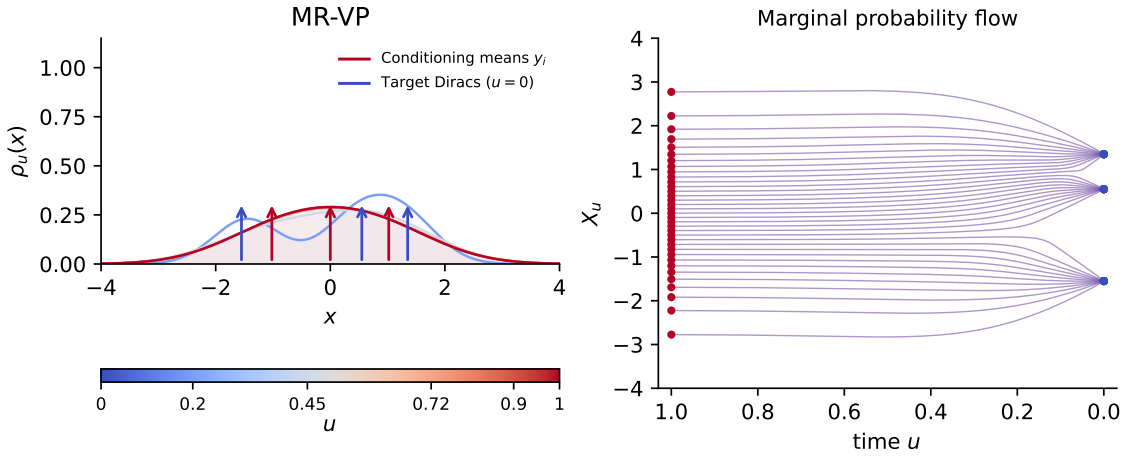
An important simplification here is the use of an isotropic covariance matrix. There are situations where the data might be better learned with non-isotropic Gaussians (Dhariwal and Nichol, 2021), or non-diagonal (Xia *et al.*, 2026; Voleti *et al.*, 2022). Such extra generalisations are beyond the topic of this thesis, but remain important to acknowledge.

The diffusion construction above uses a Gaussian terminal distribution. This choice can be restrictive for tasks whose source distribution is itself a data distribution. There are many tasks that are better described as finding a coupling between two data distributions. Some examples include image super resolution, speech noise suppression, and TTS.

Although Grad-TTS has shown success for diffusion, the framework is misaligned and becomes hard to work with. Grad-TTS simplifies the terminal distribution to a Gaussian with unit variance, which may be artificially increasing the difficulty of the problem. Move paragraph.



(a) Common conditioning mean $Y = 0$: standard-normal source



(b) Paired conditioning points y_i: Gaussian-mixture source

Figure 3.2: One-dimensional MR-VP paths for the same three target Diracs. Left: forward densities; right: marginal probability-flow trajectories towards the data at $u = 0$. In (a), $Y = 0$ for every data point and the source approaches $\mathcal{N}(0, 1)$. In (b), each target point x_i is paired with a fixed conditioning point y_i, with conditional law $Y \mid X = x_i \sim \delta_{y_i}$. The source approaches $\frac{1}{3}\sum_i \mathcal{N}(y_i, 1)$. The conditioning Diracs specify the reversion means; the source itself is a Gaussian mixture. With the finite noise schedule shown, these source distributions are approximate.

There are methods better suited to such problems. These methods are motivated by optimal transport theory. Specifically they are concerned with dynamic OT, which aims to find the minimum energy vector field that describes a transport between two distributions.

3.2.4 Schrödinger bridges

The original SB problem describes paths between two observations at $u = 0$ and $u = 1$. The paths are required to have specified endpoint marginals and are compared with a reference path law P_{ref}. The Schrödinger bridge minimises path-space Kullback–Leibler divergence subject to those marginal constraints:

$$\min_{P \in \mathcal{P}_{[0,1]}(p_0,p_1)} D_{\text{KL}}(P \| P_{\text{ref}}) \tag{3.15}$$

Under standard regularity conditions, the solution can be represented using positive Schrödinger potentials Ψ_u and $\hat{\Psi}_u$ in the forward and reverse SDEs

$$\mathrm{d}\mathbf{X} \coloneqq \left[\mathbf{F}(\mathbf{X}, u) + g^2(u)\nabla \log \mathbf{\Psi}_u\right] \mathrm{d}u + g(u)\mathrm{d}\mathbf{W}_u \tag{3.16}$$

$$\mathrm{d}\mathbf{X} \coloneqq \left[\mathbf{F}(\mathbf{X}, u) - g^2(u)\nabla \log \hat{\mathbf{\Psi}}_u\right] \mathrm{d}u + g(u)\mathrm{d}\hat{\mathbf{W}}_u \tag{3.17}$$

The time-u marginal density is proportional to the product $\hat{\Psi}_u \Psi_u$; the proportionality constant is fixed by normalisation. Assume linear drift $\mathbf{F}(\mathbf{X}) = f(u)\mathbf{X}$ and Gaussian boundaries

$$p_0 = \mathcal{N}(\mathbf{X}, \epsilon_0^2 \mathbf{I}) \quad p_1 = \mathcal{N}(\mathbf{Y}, \epsilon_1^2 \mathbf{I}) \tag{3.18}$$

where $\epsilon_1 = e^{\int_0^1 f(u)\mathrm{d}u}\epsilon_0$. As $\epsilon_0 \to 0$, the closed-form solution (Bunne *et al.*, 2023) is expressed as

$$\hat{\mathbf{\Psi}}_u = \mathcal{N}\left(e^{\int_0^u f(s)\mathrm{d}s}\mathbf{X}, e^{2\int_0^u f(s)\mathrm{d}s}\int_0^u \frac{g^2(s)}{e^{2\int_0^s f(v)\mathrm{d}v}}\mathrm{d}s\mathbf{I}\right) \tag{3.19}$$

$$\mathbf{\Psi}_u = \mathcal{N}\left(\frac{e^{\int_0^u f(s)\mathrm{d}s}}{e^{\int_0^1 f(s)\mathrm{d}s}}\mathbf{Y}, e^{2\int_0^u f(s)\mathrm{d}s}\left[\int_0^1 \frac{g^2(s)}{e^{2\int_0^s f(v)\mathrm{d}v}}\mathrm{d}s - \int_0^u \frac{g^2(s)}{e^{2\int_0^s f(v)\mathrm{d}v}}\mathrm{d}s\right]\mathbf{I}\right) \tag{3.20}$$

Given that the forward $\mathbf{\Psi}$ is guaranteed to have mean $\mathbf{Y}$ at $u = 1$, and the reverse is guaranteed to have mean $\mathbf{X}$ at $u = 0$, the solution can be simplified by considering reference paths with zero drift $f(u) = 0$, leading to

$$\hat{\mathbf{\Psi}}_u = \mathcal{N}\left(\mathbf{X}, \int_0^u g^2(s)\mathrm{d}s\mathbf{I}\right) \tag{3.21}$$

$$\mathbf{\Psi}_u = \mathcal{N}\left(\mathbf{Y}, \int_0^1 g^2(s)\mathrm{d}s - \int_0^u g^2(s)\mathrm{d}s\mathbf{I}\right) \tag{3.22}$$

the mean and covariance of the solution p are

$$\mathbf{M}(\mathbf{X}, \mathbf{Y}, u) = \left(1 - \frac{\int_0^u g^2(s)\mathrm{d}s}{\int_0^1 g^2(s)\mathrm{d}s}\right)\mathbf{X} + \left(\frac{\int_0^u g^2(s)\mathrm{d}s}{\int_0^1 g^2(s)\mathrm{d}s}\right)\mathbf{Y} \tag{3.23}$$

$$\mathbf{\Sigma} = \int_0^u g^2(s)\mathrm{d}s\left(1 - \frac{\int_0^u g^2(s)\mathrm{d}s}{\int_0^1 g^2(s)\mathrm{d}s}\right)\mathbf{I} \tag{3.24}$$

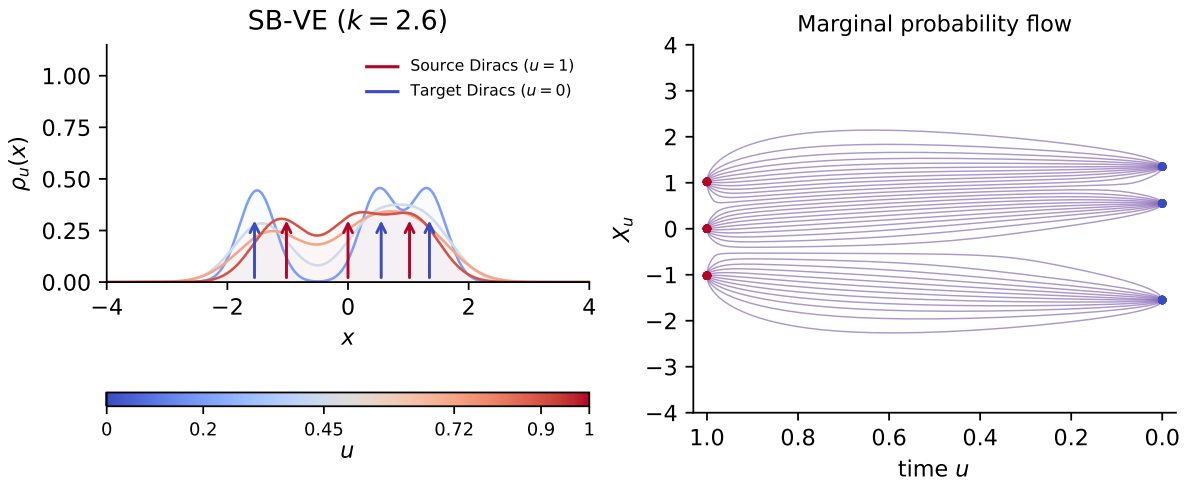


Figure 3.3: Variance-exploding bridge with $k = 2.6$. The conditional variance vanishes at both endpoints.

This solution is related to the diffusion framework described in section 3.2. Bridge paths can connect prescribed endpoint distributions, and this Gaussian example has zero conditional variance at its boundaries. This property alone does not establish lower prediction error or better generalisation than a mean-reverting SDE.

An important distinction with supervised learning is that samples from the boundaries are paired

$$(\mathbf{X}^{(i)}, \mathbf{Y}^{(i)}) \sim \pi(\mathbf{X}, \mathbf{Y}) \tag{3.25}$$

3.2.5 Variance-exploding bridge

Consider a purely variance-exploding reference process

$$\mathrm{d}\mathbf{X} := \mathbf{0}\mathrm{d}u + \sqrt{ck^{2u}}\mathrm{d}\mathbf{W}_u \tag{3.26}$$

$$\sigma_u^2 := \frac{c(k^{2u} - 1)}{2\log k}, \tag{3.27}$$

$$\mathbf{M}_u(\mathbf{X}_0, \mathbf{Y}) = \left(1 - \frac{\sigma_u^2}{\sigma_1^2}\right)\mathbf{X}_0 + \frac{\sigma_u^2}{\sigma_1^2}\mathbf{Y}, \tag{3.28}$$

$$\Sigma(u) = \sigma_u^2\left(1 - \frac{\sigma_u^2}{\sigma_1^2}\right)\mathbf{I}, \tag{3.29}$$

3.2.6 Path variance and flow matching

As illustrated in figure 3.4, for the variance-exploding bridge the mean $\mathbf{M}$ tends to linear interpolation as $k \to 1$. The variance Σ is 0 at the boundaries and increases towards the centre. This dynamic movement can make deep learning more difficult. Given that this variance is artificial, it is important to determine the benefits of this noise.

For conditional flow matching, the path considered here uses a linearly interpolating mean and

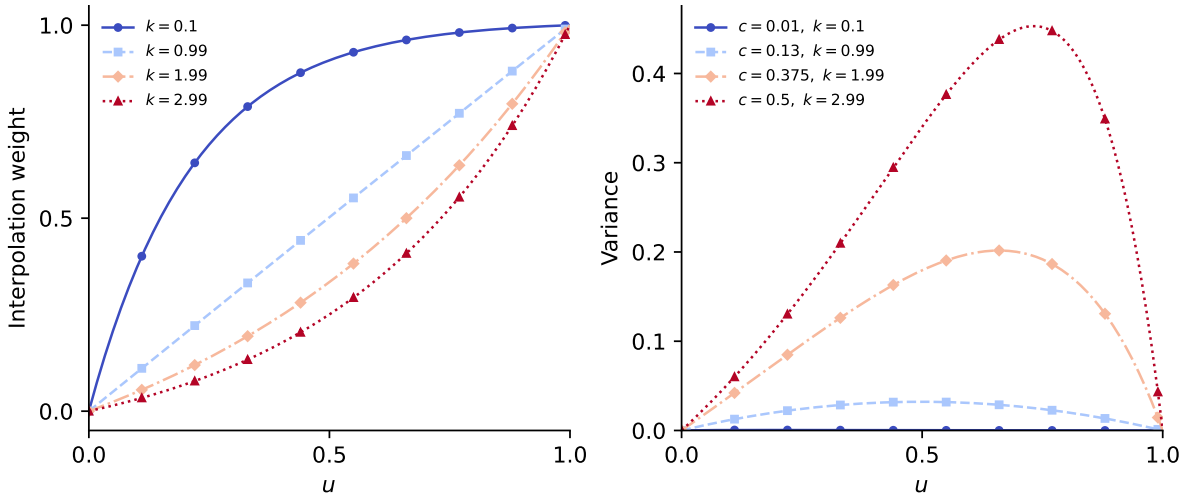


Figure 3.4: SB-VE interpolation weight and conditional variance as functions of depth u. Left: varying k with $c = 0.4$; the normalised weight is independent of c and approaches linear interpolation as $k \to 1$. Right: the displayed pairs of c and k vary together. Variance vanishes at both endpoints; at fixed k, it scales linearly with c.

	SB-VE	ICFM
Exact interpolation	y	n
Static variance	n	y
Static gradient	n	y

Table 3.1: Comparison of the SB-VE and ICFM conditional paths. Exact interpolation refers to unsmoothed endpoints; static gradient refers to the conditional mean velocity.

constant covariance. Flow matching learns the corresponding velocity field using the objective in section 3.3.3.

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathbf{M}_u = \mathbf{Y} - \mathbf{X}_0 \tag{3.30}$$

The conditional mean follows a straight path, with $\frac{\mathrm{d}}{\mathrm{d}u}\mathbf{M}_u = \boldsymbol{Y} - \mathbf{X}_0$. The target $\mathbf{X}_0 - \boldsymbol{Y}$ below is therefore the velocity in the reverse depth direction (from $u = 1$ towards $u = 0$), which is the convention used for direct prediction. This is McCann's displacement interpolation only in the zero-noise limit and when the endpoint coupling is optimal for the transport cost. The conditional path is described as

$$\mathbf{M}(\mathbf{X}_0, \mathbf{Y}, u) = u\mathbf{Y} + (1-u)\mathbf{X}_0 \quad \Sigma(u) = c\mathbf{I} \tag{3.31}$$

For $c > 0$, the conditional endpoints retain covariance $c\mathbf{I}$ rather than being exact point masses.

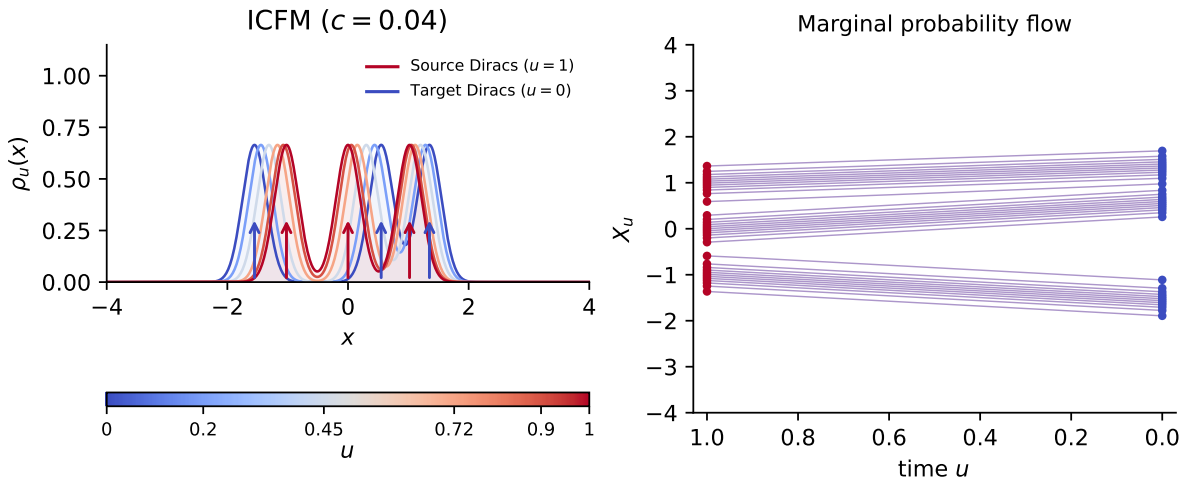


Figure 3.5: Conditional path for flow matching: linear interpolation with constant covariance $0.04\mathbf{I}$, including at the endpoints.

3.3 Simulation-free learning objectives

3.3.1 Score and clean-data prediction

The marginal score is typically intractable, but for a Gaussian transition the conditional score $\nabla \log p_{u|0}(\mathbf{X}_u|\mathbf{X}_0)$ is available in closed form. Although this target depends on $\mathbf{X}_0$, a network that receives only $(\mathbf{X}_u, u)$ can learn the marginal score by conditional score matching. With an unrestricted function class and samples from the stated training distribution, the population regression optimum is the marginal score (Song *et al.*, 2021b). The objective is

$$\boldsymbol{\theta}^* \in \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{\mathrm{u}\sim\mathcal{U}(0,1),\mathbf{X}_0\sim\hat{p}_{\mathrm{data}},\mathbf{X}_\mathrm{u}\sim p_{\mathrm{u}|0}(\cdot|\mathbf{X}_0)} \left\|\mathbf{S}_{\boldsymbol{\theta}}(\mathbf{X}_\mathrm{u}, \mathrm{u}) - \nabla_{\mathbf{X}_\mathrm{u}} \log p_{\mathrm{u}|0}(\mathbf{X}_\mathrm{u}|\mathbf{X}_0)\right\|_2^2. \tag{3.32}$$

Here $\mathrm{u} \sim \mathcal{U}(0,1)$ is the random depth, $\mathbf{X}_0$ is sampled from the empirical data distribution, and $\mathbf{X}_\mathrm{u}$ is sampled from its conditional transition law. By sampling $\mathbf{X}_1 \sim p_1$ and integrating the probability-flow ODE backwards with the estimated marginal score, data samples can be generated. Note how depth is optimised at the "infinite layer limit", and a random "layer" is optimised per sample per optimisation step.

In the case where the SDE has a closed-form Gaussian solution, the conditional score becomes easy to compute. By assuming that the conditional probability path is Gaussian

$$p_{u|0}(\mathbf{X}_u|\mathbf{X}_0) = \mathcal{N}(\mathbf{X}_u; \mathbf{M}(\mathbf{X}_0, u), \Sigma(u)) \tag{3.33}$$

where $\mathbf{M}$ is the mean and Σ is the covariance. Write $\mathbf{X}_u = \mathbf{M}(\mathbf{X}_0, u) + \Sigma(u)^{1/2}\mathbf{Z}$, where $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The conditional score is then

$$\nabla_{\mathbf{X}_u} \log p_{u|0}(\mathbf{X}_u|\mathbf{X}_0) = -\Sigma(u)^{-\frac{1}{2}}\mathbf{Z}. \tag{3.34}$$

This form gives a simple training target for the score estimator.

An alternative to predicting the score is to predict the clean data given the noisy sample:

$$\boldsymbol{\theta}^* \in \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{\mathrm{u}\sim\mathcal{U}(0,1),\, \mathbf{X}_0\sim\hat{p}_{\mathrm{data}},\, \mathbf{X}_{\mathrm{u}}\sim p_{\mathrm{u}|0}(\cdot|\mathbf{X}_0)} \left\|\mathbf{D}_{\boldsymbol{\theta}}(\mathbf{X}_{\mathrm{u}}, \mathrm{u}) - \mathbf{X}_0\right\|_2^2 . \tag{3.35}$$

For the special case $\mathbf{X}_u = \mathbf{X}_0 + u\mathbf{Z}$, with $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the optimal denoiser satisfies

$$\nabla_{\mathbf{X}_u} \log p_u(\mathbf{X}_u) = \frac{\mathbf{D}^*_{\boldsymbol{\theta}}(\mathbf{X}_u, u) - \mathbf{X}_u}{u^2}. \tag{3.36}$$

For the general affine Gaussian transition $\mathbf{X}_u = a_u\mathbf{X}_0 + b_u + \Sigma(u)^{1/2}\mathbf{Z}$, the corresponding identity is

$$\nabla_{\mathbf{X}_u} \log p_u(\mathbf{X}_u) = \Sigma(u)^{-1} \left(a_u\mathbf{D}^*_{\boldsymbol{\theta}}(\mathbf{X}_u, u) + b_u - \mathbf{X}_u\right). \tag{3.37}$$

These identities apply to the population-optimal denoiser; a trained network only approximates that optimum. The denoiser objective predicts the clean target at every noise level (Karras *et al.*, 2022).

Simulation-free training is analogous to teacher-forcing. An instantaneous point from a reference path is sampled, and input to the estimator. A sufficiently trained model will align with the reference path, much like a sequence model aligning to likely sequences.

There are many ways to develop an architecture for score estimators $\mathbf{S}_{\boldsymbol{\theta}}$. An identifying feature of a score estimator is that it depends on the depth variable u in addition to the current sample $\mathbf{X}_u$. Two popular examples are NCSN++ and DiT.

An NCSN++ is a modified U-Net (Ronneberger *et al.*, 2015) with a learnt u-step embedding applied to each Resnet block. The U-Net remains similar to the original implementation as a series of downsampling layers, a bottleneck, and then a series of upsampling layers. As with all convolutional neural networks, contextual dependence can only be modelled within the kernels receptive field.

A more modern approach is to use a DiT. A DiT uses the transformer architecture with adaptive layer-norm to condition on the u-step. Since transformers are sequence models, modalities such as images require tokenisation. A strong method for this is to segment the image into patches and compute a patch embedding for each patch.

Both the NCSN++ and DiT have a dimensionality-reduction step for features such as images and spectrograms. The NCSN++ uses down-sampling layers and DiT uses patch embeddings. Such steps are interesting as the noise typically exists on the highest dimensional manifold whereas the data may lie on a much lower manifold e.g. an image of a blue sphere has much less variance than a standard normal sample. It might be the case that dimensionality reduction offers a denoising bias as it forces local correlation during the learned reverse process which is beneficial for a data-generating score network.

Path	Prediction target	Inference
Diffusion	Score or clean data	Probability-flow ODE
Variance-exploding bridge	Clean endpoint	Bridge sampler or direct prediction
Flow matching (linear path)	Reverse velocity	ODE or direct prediction

Table 3.2: Path, prediction, and inference choices considered in this chapter.

3.3.2 Bridge data prediction

For this example, a data-prediction objective regresses the clean endpoint from a path sample:

$$\mathcal{L}_{\mathrm{DP}}(\theta) = \mathbb{E}_{(\mathbf{X}_0,\mathbf{Y})\sim\pi,\,\mathrm{u}\sim U(0,1),\,\mathbf{X}_\mathrm{u}\sim p_{\mathrm{u}|0,Y}(\cdot|\mathbf{X}_0,\mathbf{Y})}\left[\left\|\mathbf{D}_\theta(\mathbf{X}_\mathrm{u} \mid \boldsymbol{Y},\mathrm{u}) - \mathbf{X}_0\right\|^2\right] \tag{3.38}$$

3.3.3 Flow matching by velocity prediction

For the linear interpolation path in section 3.2.6, conditional flow matching regresses the reverse-depth velocity:

$$\begin{aligned}\boldsymbol{\theta}^* &= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{(\mathbf{X}_0,\mathbf{Y})\sim\pi,\,\mathrm{u}\sim U(0,1),\,\mathbf{Z}\sim\mathcal{N}(0,I)}\left[\left\|\mathbf{F}_{\boldsymbol{\theta}}(\mathbf{X}_\mathrm{u} \mid \mathbf{Y},\mathrm{u}) - (\mathbf{X}_0 - \mathbf{Y})\right\|^2\right], \\ \mathbf{X}_\mathrm{u} &= \mathbf{M}(\mathbf{X}_0,\mathbf{Y},\mathrm{u}) + \sqrt{c}\mathbf{Z}\end{aligned} \tag{3.39}$$

3.4 Conditional generation and speech data

An extension of the generative modelling task is conditional generation. This involves generating realistic data samples that are likely given a conditioning signal e.g. a text prompt. The notion of conditional distribution between two random variables becomes relevant here.

$$p(\mathbf{X}|\boldsymbol{y}) = ? \tag{3.40}$$

There are several ways to influence the generation process. One approach is to form a Bayesian score

$$\nabla \log p_u(\mathbf{X}_u|\mathbf{y}) = \nabla \log p_u(\mathbf{X}_u) + \nabla \log p_u(\mathbf{y}|\mathbf{X}_u) \tag{3.41}$$

$p_u(\mathbf{y}|\mathbf{X}_u)$ is known as the *classifier*. Classifier-guided generation depends heavily on the quality of the classifier, which adds additional compute cost and often fails to generate high-quality samples. Two advanced techniques considered in this thesis are classifier-free guidance and mean-reverting SDEs.

3.4.1 Classifier-free guidance

One option is to provide conditioning labels during training to optimise $\nabla \log p_u(\mathbf{X}_u|\mathbf{y})$, e.g. the score network could use cross-attention to be informed by the conditioning signal. In doing so, the score estimator is at risk of overfitting to the labelled data and no longer providing a diverse density possible with the unconditional model.

1	Sample length $\mathrm{l} \sim p_l(\mathrm{l})$
2	Sample terminal distribution $\hat{p}_1 \sim \mathcal{W}_\mathrm{l}$
3	Sample data $\hat{\mathbf{X}} \sim p_{\text{data}}$
4	Define initial $\hat{p}_0 = \mathcal{N}(\hat{\mathbf{X}}, \Sigma_0)$
5	Sample path $p \sim \mathcal{P}_{[0,1]}$ with boundaries $\hat{p}_0$ and $\hat{p}_1$
6	Single step MC $\mathbb{E}_{\mathbf{X}_0 \sim p_0}$

Table 3.3: Algorithm for variable length generation

Classifier-free guidance overcomes this by training a conditional-unconditional mixture. The "classifier" can be written as

$$\nabla_{\mathbf{X}_u} \log p_u(\mathbf{y}|\mathbf{X}_u) = \nabla_{\mathbf{X}_u} \log p_u(\mathbf{X}_u|\mathbf{y}) - \nabla_{\mathbf{X}_u} \log p_u(\mathbf{X}_u). \tag{3.42}$$

Define ω to control the mixture of conditional and unconditional scores. The guided score is

$$\nabla \log p_u(\mathbf{X}_u|\mathbf{y}, \omega) \coloneqq \nabla_{\mathbf{X}_u} \log p_u(\mathbf{X}_u) + \omega\big(\nabla_{\mathbf{X}_u} \log p_u(\mathbf{X}_u|\mathbf{y}) - \nabla_{\mathbf{X}_u} \log p_u(\mathbf{X}_u)\big) \tag{3.43}$$

$$= \omega \nabla \log p_u(\mathbf{X}_u|\mathbf{y}) + (1-\omega)\nabla \log p_u(\mathbf{X}_u) \tag{3.44}$$

At $\omega = 0$ this is the unconditional score; at $\omega = 1$ it is the conditional score. Values $\omega > 1$ extrapolate in the direction of the classifier gradient and give stronger guidance in the score field. This linear combination does not, by itself, establish the distribution produced by a particular sampler. The conditional and unconditional scores can be trained together by randomly replacing $\mathbf{y}$ with the null label $\oslash$ during training.

3.4.2 Mean reversion with conditional score

In some cases, a coarse representation of the data is predictable, and diffusion is then used to refine the features. Here, the predicted coarse representation defines the mean of the terminal distribution

$$\mathbf{Y} \coloneqq \mathbf{E}_{\boldsymbol{\phi}}(\mathbf{y}). \tag{3.45}$$

where $\mathbf{E}_{\boldsymbol{\phi}}$ is a deterministic encoder and $\mathbf{Y}$ is the mean of p_1 conditional on $\mathbf{y}$. Then for any SDE with a mean reverting drift e.g. $\mathbf{F}(u) = \mathbf{Y} - \mathbf{X}_u$ the score estimator is constrained to flows that revert to the mean $\mathbf{Y}$.

3.4.3 Variable-length speech

Some data types such as speech have an associated length l with each sample $\mathbf{X}_0 \in \mathbb{R}^{d \times \mathrm{l}}$, where d is the static dimension of the data e.g. number of Mel-spectrogram channels. This means the sampled terminal distribution $\hat{p}_1$ must also sample or predict a length to operate in. The length of a sample remains constantly for any u.

The constraints

$$p_0(\mathbf{X}_0) = \int p(\mathbf{X}) \mathrm{d}\mathcal{P} \mathrm{d}\mathcal{W}(\mathrm{l}) \mathrm{dl} \tag{3.46}$$

$$p_0(\mathbf{X}_0) = p_L(\mathrm{l}) \int \int \mathrm{p}_{u=0}(\mathbf{X}_0) \mathrm{d}\mathcal{P}_{\mathrm{p}_1 \# \mathbf{X}_0}(\mathrm{p}_u) \mathrm{d}\mathcal{W}(\mathrm{p}_1) \tag{3.47}$$

$$p_0(\mathbf{X}_0) = \int \int \int p_0(\mathbf{X}_0|\mathbf{X}_1) \mathcal{N}(\mathrm{d}\mathbf{X}_1; \mathbf{Y}, \Sigma) \mathcal{W}_{\mathbb{R}^{d\times 1}}(\mathrm{d}\mathbf{Y}, \mathrm{d}\Sigma) p_l(\mathrm{dl}) \tag{3.48}$$

$$p(\mathbf{X}_0|\mathbf{y}) = \int_{\mathbb{Z}_+} \iint_{\mathbb{R}^{d\times 1}} p(\mathbf{X}_0|\mathbf{X}_1, \mathbf{Y}) p(\mathbf{X}_1|\mathbf{Y}) p(\mathbf{Y}|\mathbf{y}, \mathrm{l}) p(\mathrm{l}|\mathbf{y}) \mathrm{d}\mathbf{X}_1 \mathrm{d}\mathbf{Y} \mathrm{dl} \tag{3.49}$$

3.4.4 Paired endpoints for noise suppression

Noisy speech is defined as a clean signal $\mathbf{X}$ convolved with a room impulse response $\boldsymbol{H}$ and additive noise $\mathbf{Z}$

$$\mathbf{Y} \coloneqq \mathbf{X} * \boldsymbol{H} + \mathbf{Z} \tag{3.50}$$

The goal of noise suppression is to retrieve $\mathbf{X}$ given $\mathbf{Y}$.

$$q_1(\mathbf{Y}) = \int p_z(\mathbf{Y} - \mathbf{X} * \boldsymbol{H}) q_0(\mathrm{d}\mathbf{X}) \tag{3.51}$$

The paired distributions motivate a conditional path. Instead of a variance preserving SDE, consider a variance exploding SDE

$$\mathrm{d}\mathbf{X} \coloneqq \gamma(\mathbf{Y} - \mathbf{X}_u)\mathrm{d}u + \sqrt{ck^{2u}}\mathrm{d}\mathbf{W}_u. \tag{3.52}$$

$$\mathbf{M}(\boldsymbol{Y}, \mathbf{X}, u; \gamma) = e^{-\gamma u}\mathbf{X} + (1 - e^{-\gamma u})\boldsymbol{Y} \qquad \Sigma(u; \gamma, c, k) = \frac{c(k^{2u} - e^{-2\gamma u})}{2(\gamma + \log k)}\mathbf{I} \tag{3.53}$$

When $\boldsymbol{H} = I$, the flow-matching target also equals the additive-noise residual. With reverberation, this residual is $\mathbf{Y} - \mathbf{X} = \mathbf{X} * \boldsymbol{H} - \mathbf{X} + \mathbf{Z}$, so it is not the background noise alone:

$$\mathbf{Y} - \mathbf{X} = \mathbf{X} * \boldsymbol{H} - \mathbf{X} + \mathbf{Z} \tag{3.54}$$

3.5 Inference and likelihood evaluation

3.5.1 Probability-flow ODE

An attractive feature of CNFs is that the change of variables formula becomes easier to compute. So far the reverse process has been presented in SDE form, but for generation it is possible to use an equivalent ODE

$$\mathbf{H}_{\boldsymbol{\theta}}(\mathbf{X}, u) \coloneqq \mathbf{F}(\mathbf{X}, u) - \frac{1}{2} g(u)^2 \mathbf{S}_{\boldsymbol{\theta}}(\mathbf{X}, u) \tag{3.55}$$

This vector field defines the probability-flow ODE. For the variance preserving example, the probability-flow ODE is

$$\mathbf{H}(\mathbf{X}, u) \coloneqq \frac{1}{2}\left(\boldsymbol{Y} - \mathbf{X} - \mathbf{S}_{\boldsymbol{\theta}}(\mathbf{X}, u)\right)\beta(u). \tag{3.56}$$

For classifier-free guidance, apply the following probability flow ODE

$$\mathrm{d}\mathbf{X} = \frac{1}{2}\left((\boldsymbol{Y} - \mathbf{X}_u) - \omega \boldsymbol{S}_\theta(\mathbf{X}_u, u|\boldsymbol{y}) + (\omega - 1)\boldsymbol{S}_\theta(\mathbf{X}_u, u|\oslash)\right)\beta(u)\mathrm{d}u. \tag{3.57}$$

For a conditional mean reversion, note how $\mathbf{Y}$ is a random variable of the conditioning $\mathbf{y}$, unlike a global $\boldsymbol{Y}$ left as a design choice in equation 3.57. This represents a key difference between image and speech generative models seen later.

3.5.2 Numerical sampling

To retrieve data samples, the following reverse initial value problem must be solved with black-box ODE solvers

$$\mathbf{X}_0 = \mathbf{X}_1 + \int_1^0 \mathbf{H}_{\boldsymbol{\theta}}(\mathbf{X}_u, u)\mathrm{d}u. \tag{3.58}$$

As stated earlier, this form of CNF is a probability-flow ODE. The lack of diffusion is useful as it constrains the input variance of the neural estimator $\mathbf{S}_{\boldsymbol{\theta}}$.

A simple method is the Euler sampler

$$\mathbf{X}_{u-h} = \mathbf{X}_u - \mathbf{H}_{\boldsymbol{\theta}}(\mathbf{X}_u, u)h. \tag{3.59}$$

where $h \leq 1$ is the step size. The number of steps required is $\frac{1}{h}$. This method has a fixed number of steps per solve. Typically, more steps improve accuracy, presenting a compute-accuracy trade-off.

There are many options for ODE solvers. For example, the fourth-order Runge-Kutta method evaluates the vector-field four times per step (Runge, 1895; Kutta, 1901). Adaptive methods dynamically increase or decrease the step size under select criteria such as an error tolerance between fourth and fifth order evaluations (Dormand and Prince, 1980). The development of diffusion models has lead to customised solvers for the generative task e.g. with special weighting schemes (Lu *et al.*, 2022).

A key area of research here is improving quality with less steps.

3.5.3 Bridge sampling and direct data prediction

The ODE sampler that solves the SB-VE problem is defined as

$$\begin{aligned}\boldsymbol{X}_{u_{i-1}} = &\frac{\sigma_{u_{i-1}}\sqrt{\sigma_1^2-\sigma_{u_{i-1}}^2}}{\sigma_{u_i}\sqrt{\sigma_1^2-\sigma_{u_i}^2}}\boldsymbol{X}_{u_i} \\ &+\frac{1}{\sigma_1^2}\left(\sigma_1^2-\sigma_{u_{i-1}}^2-\frac{\sqrt{\sigma_1^2-\sigma_{u_i}^2}\,\sigma_{u_{i-1}}\sqrt{\sigma_1^2-\sigma_{u_{i-1}}^2}}{\sigma_{u_i}}\right)\boldsymbol{D}_{\boldsymbol{\theta}}(\boldsymbol{X}_{u_i},\boldsymbol{Y},u_i) \\ &+\frac{1}{\sigma_1^2}\left(\sigma_{u_{i-1}}^2-\frac{\sigma_{u_i}\sigma_{u_{i-1}}\sqrt{\sigma_1^2-\sigma_{u_{i-1}}^2}}{\sqrt{\sigma_1^2-\sigma_{u_i}^2}}\right)\boldsymbol{Y}.\end{aligned} \tag{3.60}$$

for an increasing sequence of u_i. The denoiser is evaluated at the current state depth u_i. The formula is singular at the endpoint $u_i = 1$, so an implementation needs an endpoint special case or a limiting update.

An alternative to the SB-ODE proposed in this thesis is *direct data prediction*. In section 3.3 the data prediction objective is used to make the estimator well conditioned to clean data across all noise levels. An added benefit of the data prediction loss is that the estimator becomes a data predictor, and may be a sufficiently strong model without the need to solve an ODE. Here the term "direct data prediction" is used to describe the use of a noise conditional model as a data predictor instead of an ODE.

Direct data prediction with data prediction loss is

$$\boldsymbol{X}_0 = \boldsymbol{D}_{\boldsymbol{\theta}}(\boldsymbol{Y},\boldsymbol{Y},1) \tag{3.61}$$

and for flow-matching loss

$$\boldsymbol{X}_0 = \boldsymbol{F}_{\boldsymbol{\theta}}(\boldsymbol{Y},\boldsymbol{Y},1)+\boldsymbol{Y} \tag{3.62}$$

Direct prediction evaluates the network once at the source depth. It does not integrate the intermediate path. For a path with nonzero endpoint variance, evaluation at the source mean should be distinguished from sampling the full source distribution.

3.5.4 Point-cloud illustrations of inference

Figure 3.6 illustrates discrete inference trajectories for paired endpoints using MR-VP, a Schrödinger bridge, and flow matching. These examples complement the opening probability-path overview by showing how the samplers move individual points.

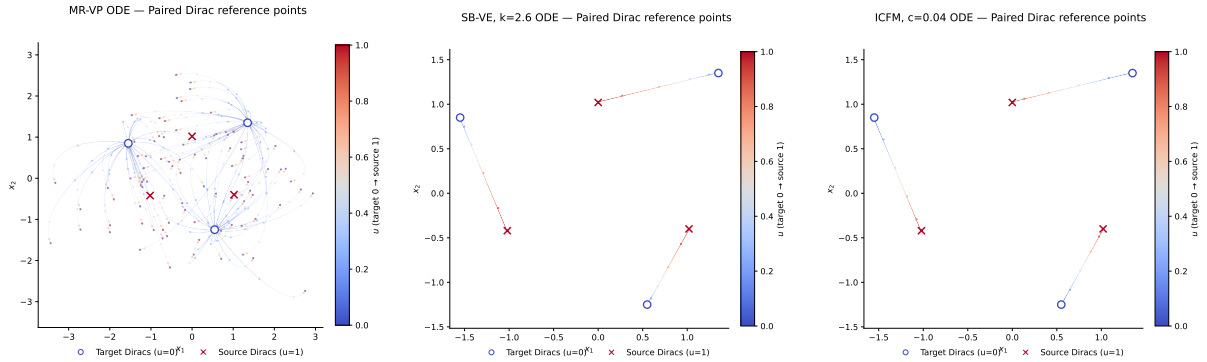


Figure 3.6: Point-cloud inference for paired three-point distributions: MR-VP (left), SB-VE with $k = 2.6$ (centre), and ICFM with $c = 0.04$ (right). Colours indicate depth, with generation proceeding towards $u = 0$. Analytic conditional predictors and 100 discrete integration steps are used. Each source point uniquely identifies its target, so trajectories with identical starts coincide. MR-VP starts from Gaussian noise about the conditioning mean; SB and ICFM start at the observed source point. SB integration begins at $u = 1 - 10^{-4}$ to avoid singular endpoint coefficients.

3.5.5 Likelihood evaluation

The likelihood along the path is

$$\log p_0(\mathbf{X}_0) = \log p_1(\mathbf{X}_1) + \int_0^1 \nabla_{\mathbf{X}_u} \cdot \mathbf{H}_{\boldsymbol{\theta}}(\mathbf{X}_u, u) \mathrm{d}u. \tag{3.63}$$

The divergence can evaluated with a trace estimator (Skilling, 1989; Hutchinson, 1989)

$$\nabla_{\mathbf{X}} \cdot \mathbf{H}_{\boldsymbol{\theta}}(\mathbf{X}, u) = \mathbb{E}_{\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\mathbf{Z}^\top \nabla_{\mathbf{X}} \mathbf{H}_{\boldsymbol{\theta}}(\mathbf{X}, u) \mathbf{Z} \right]. \tag{3.64}$$

Although score-based modelling is simulation-free, it removes the ability to maximise likelihood directly. Despite this, the learnt density outperforms non-continuous-depth alternatives (Song *et al.*, 2021b). Interestingly, there are methods for maximum likelihood training Song *et al.* (2021a), yet these do not produce higher quality samples. Such research emphasises the fact that evaluating generative models is a non-trivial task.

A computable likelihood allows the model to be robustly evaluated. Despite this advantage, evaluating likelihood on practical tasks such as controllable image and speech generation is not commonplace. The following empirical chapter explores this gap.

3.6 Summary

Simulation-free training combines a prescribed probability path with an accessible conditional regression target. Diffusion, bridge, and linear flow-matching path constructions provide different endpoint and variance choices. Score, clean-data, and velocity prediction connect these paths to inference procedures, while conditioning supports generation and paired speech enhancement.

The next chapter evaluates likelihood in practical diffusion models. The subsequent noise-suppression chapter examines paired paths, variance choices, and direct prediction.

Chapter 4

An empirical evaluation of log-likelihood

This chapter investigates the properties of diffusion-model likelihood in practical implementations. The models considered are SDXL for text-to-image generation and Grad-TTS for speech synthesis. These models use different probability paths and conditioning mechanisms.

Differences between the original model formulations and practical implementations may affect likelihood. For example, the probability-flow ODE used with SDXL contains a guidance-scale parameter. Does the evaluated density shift towards images depicting the caption as that scale increases? Grad-TTS also uses an utterance-dependent probability flow conditioned on $\mathbf{y}$. How much does the model rely on the encoded mean $\mathbf{Y}$, and how much does it represent linguistic structure?

The above questions are answered through a series of experiments. First the likelihood of a sample through a probability path is measured to confirm that likelihood increases during the generative process. The guidance scale is also varied to verify if the density changes. Then, special evaluations are taken to measure expected behaviour for the task. These tasks apply the density model for classification tasks e.g. which transcript fits a spoken utterance the best.

The measured likelihood generally increases as $u \to 0$, but at different rates: Grad-TTS changes most in the final quarter, whereas SDXL likelihood is not monotone. In the reported CFG comparison, increasing the guidance scale is associated with higher likelihood.

The rest of the evaluations suggest that the densities are not task-orientated. For example, using likelihood for ranking hypothesis transcriptions for speech recognition does not find the correct transcription. Given a list of confounding prompts, image likelihood struggles to find the correct prompt for an image. Such behaviours are unexpected for a high-quality density model.

The rest of this chapter outlines the models used and data they are tested on. Then details of the evaluation metrics are given. Then the experimental results are presented and discussed. The chapter ends with a final summary paragraph tying everything learned together so far.



Figure 4.1: Enter Caption

4.1 Models

SDXL is a latent diffusion model that operates in the latent space of an image autoencoder. Generation uses classifier-free guidance (CFG). The score network is a U-Net with cross-attention to text representations. The reported density is a density over latents; it is not an exact image-space likelihood unless the image transform and its Jacobian are included. The training-data statement and the precise guidance implementation should be checked against the experimental configuration.

Grad-TTS uses a transformer encoder and U-Net diffusion decoder. The encoder predicts a coarse mel-spectrogram which is used as the mean $\mathbf{Y}$. Although the mel-spectrogram has 80 dimensions, the input dimension of the U-Net is 160 to accommodate the encoder output for conditioning. Grad-TTS is trained on LJSpeech.

4.2 Data

Speech The datasets considered are the validation sets of LJSpeech (Ito and Johnson, 2017) and TED-LIUM (Rousseau *et al.*, 2012). The LJSpeech split contains around 1 hour of audiobooks read by a US-English female speaker. The TED-LIUM split contains around 3 hours of male speakers and 1 hour of female speaker TED conference audio. To observe how likelihood changes during denoising, likelihood is measured at several values of the depth coordinate u. Bits per dimension (BPD) is $-\log_2 p(\mathbf{x})/D$, where $D = \prod_i d_i$ is the number of scalar dimensions in $\mathbf{x}$. Lower BPD means higher likelihood.

Image The datasets are PACS (Li *et al.*, 2017), a domain-generalisation dataset with photo, art-painting, cartoon, and sketch domains, and CLEVR (Johnson *et al.*, 2017), a diagnostic dataset of synthetic 3D objects in eight colours. The images are generated following Lewis *et al.* (2023), including a "red sphere" example. A single-object subset of CLEVR is used. Each class-domain or shape-colour pair is repeated 20 times. Images are resized to 512^2 or 1024^2; the number of PACS classes used in these experiments should be stated explicitly. The likelihood of SDXL latents is reported, with classifier-free guidance (CFG) used to generate PACS samples. Image-distribution distance and text-image similarity are measured using Fréchet inception distance (FID) and CLIP similarity, respectively (Table 4.4).

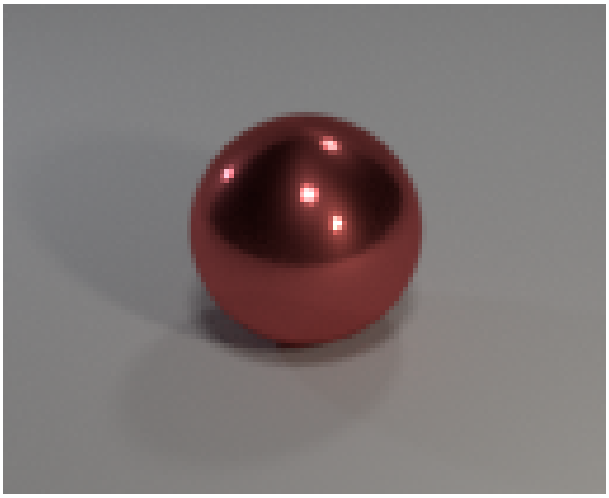


Figure 4.2: Enter Caption

4.3 Evaluation

Speech Mel-cepstral distortion (MCD) is a measure of how different two mel-cepstra (audio features) are from each other (Kominek *et al.*, 2008).[1] The root mean squared error of the log fundamental frequencies is denoted as "log-F_0 RMSE" and measures pitch/intonation accuracy.[2] Word error rate (WER) is the ratio of transcription errors to the number of words; intelligibility is measured by calculating the WER of the transcript generated by the automatic speech recognition (ASR) foundation model Whisper (Radford *et al.*, 2022). Speaker similarity with automatic speaker verification (ASV) is measured by comparing the cosine similarity between embeddings from the speaker verification foundation model Titanet (Koluguri *et al.*, 2021).

Image FID measures a distance between real and synthetic image feature distributions. CLIP similarity measures alignment between text and image representations. These metrics measure different properties from MCD and WER in the TTS experiment.

4.4 Experiments

4.4.1 Text-to-speech

To observe how likelihood changes during denoising, likelihood is measured at several values of the depth coordinate u. Bits per dimension (BPD) is $-\log_2 p(\mathbf{x})/D$, where $D = \prod_i d_i$ is the number of scalar dimensions in $\mathbf{x}$. Lower BPD means higher likelihood. Table 4.1 shows that although likelihood increases as $\mathbf{x}_1$ is denoised into data $\mathbf{x}_0$, the effect on other metrics is surprising. Likelihood is correlated with ASV, marginally with MCD, but not log-F_0 RMSE. Intelligibility decreases as likelihood improves. The results suggest that composition between the source text and hypothesis utterance is modelled by the encoder $\text{enc}(\cdot)$, and the diffusion decoder does not model the exact likelihood between text and speech $p_0(\mathbf{x}|\boldsymbol{y})$. This is verified through an ASR decoder

[1] MCD is calculated with the method presented in ESPnet (Gao *et al.*, 2023)

[2] For reference, Tacotron 2 and Fastspeech 2 achieve 0.26 and 0.24 log-F_0 RMSE respectively (Ren *et al.*, 2022).

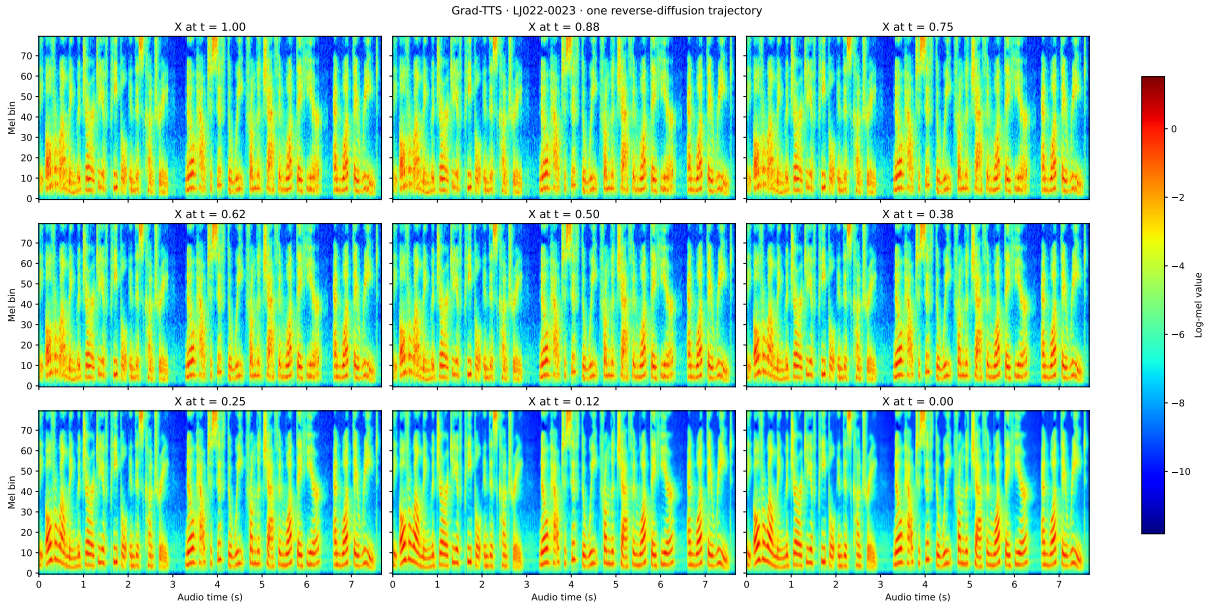


Figure 4.3: Enter Caption

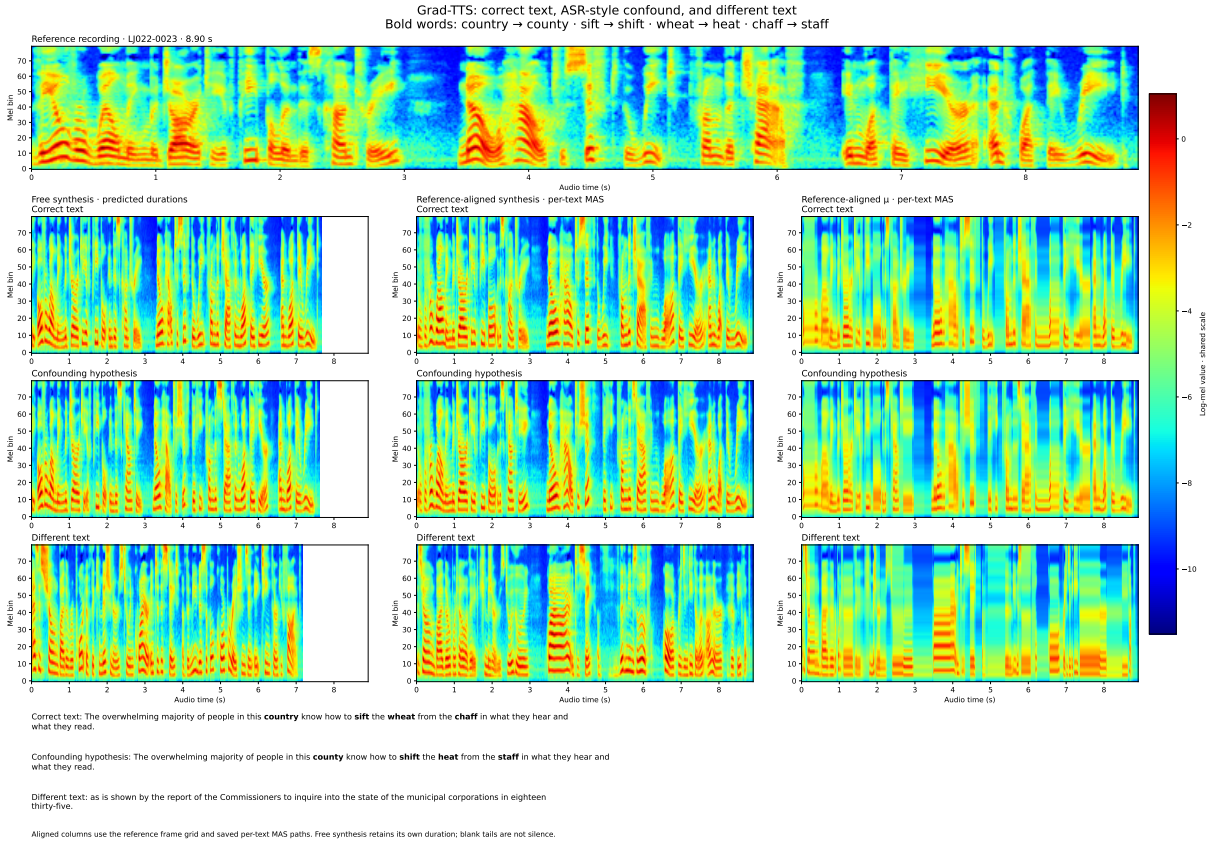


Figure 4.4: Enter Caption

Table 4.1: Quantitative assessment of Grad-TTS from depth $u = 1$ to $u = 0$.

Metric	Ref	Depth u								
		1.00	0.88	0.75	0.62	0.50	0.38	0.25	0.12	0.00
BPD ↓	-0.06	2.79	2.79	2.79	2.79	2.77	2.72	2.55	1.35	-0.08
log-F_0 RMSE ↓	0.00	0.28	0.28	0.28	0.28	0.28	0.27	0.27	0.27	0.27
MCD ↓	0.00	6.52	6.52	6.52	6.50	6.47	6.43	6.39	6.34	6.31
WER ↓	0.03	0.08	0.08	0.08	0.09	0.08	0.10	0.13	0.16	0.19
ASV ↑	1.00	0.74	0.74	0.73	0.73	0.74	0.75	0.78	0.84	0.86

rescoring experiment on LJSpeech. An ASR pipeline generates a list of hypothesis transcripts for a given utterance, which are scored by a weighted combination of an acoustic model (AM) probability and a language model probability to find the best transcription. This pipeline serves as a diagnostic tool for how well Grad-TTS likelihoods can measure speech/text compatibility. The AM probabilities are replaced with Grad-TTS likelihoods, and standard hypothesis rescoring techniques are applied (Kahn *et al.*, 2022). The models tested are a wav2vec2 model fine-tuned on 10 minutes of audiobook data (Panayotov *et al.*, 2015), and Whisper (Table 4.2).

Table 4.2: WER% ↓ of n-best list rescoring strategies including Grad-TTS.

AM Model	AM Score	Grad-TTS LL	Oracle Best	Oracle worst	Random
wav2vec2	13.3	19.6	10.5	29.0	20.2
Whisper	3.3	4.7	1.5	7.2	4.3

In all cases, the DM likelihoods yield a worse score than the AM, although better than the oracle worst and better than random in the wav2vec2 case. This suggests that the likelihoods may have some sensitivity to textual changes but lack an expressive linguistic representation. To survey how DM likelihood interacts with intelligibility and speaker quality, Grad-TTS is used to adapt TED-LIUM live-talk data (Rousseau *et al.*, 2012) to aid an ASR model trained on audiobooks. ASR models perform worse when the acoustic properties differ between train and test data. The spectrogram is passed through a Gaussian blur with kernel 5 and $\sigma = 1$, simulating text encoder output $\widetilde{\mathbf{x}} \approx \text{enc}(\boldsymbol{y})$.[3] The forward ODE is used to calculate $\mathbf{x}_1$, and $\mathbf{x}_0$ is then produced with the reverse ODE. Given that Grad-TTS is trained on LJSpeech, the newly generated $\mathbf{x}_0$ will be acoustically similar to the data the ASR model was trained on. If Grad-TTS can adapt acoustic-linguistic features, it will reduce WER. Both Euler and RK45 solvers are evaluated (Table 4.3).

The Euler method produces unintelligible audio, yet has high speaker similarity and likelihood. The RK45 sampler degrades intelligibility to a lesser degree than Euler, but speaker adaptation is negligible. The trend is that greater shift towards the LJSpeech speaker produces less intelligible audio.

Table 4.3: Quantitative assessment of cross-domain adaptation with Grad-TTS.

ODE sampler		Metric		
Forward	Reverse	WER ↓	BPD ↓	ASV ↑
None	None	0.46	1.66	0.52
Euler	Euler	0.94	-1.47	0.66
RK45	Euler	0.57	6.92	0.54
RK45	RK45	0.57	1.83	0.55



Figure 4.5: Enter Caption



Figure 4.6: Enter Caption

Table 4.4: The SDXL generative process from $u = 1$ to 0 with CFG=7 at 1024 resolution.

Metric	Data	Depth u				
		1.00	0.75	0.50	0.25	0.00
BPD ↓	-0.17	0.07	0.08	-0.08	-0.12	-0.08
FID ↓	0.00	478.91	369.91	225.89	221.40	229.08
CLIP ↑	29.50	22.72	23.49	28.10	27.97	28.16

4.4.2 Text-to-image

Similar to Table 4.1, the reported FID improves overall during the generative process, while the BPD values fluctuate. The FID remains high in this comparison; the CIFAR-10 result cited in the footnote is not directly comparable because it uses a different dataset and evaluation setup. The CLIP similarity improves over much of the trajectory, but this does not by itself show that CFG defines a more conditional likelihood than the MR-VP SDE used by Grad-TTS.

Table 4.5: Impacts of CFG, reconstruction (-R) and image sizes. *Divide in two*

Metric	Data	CFG				Size-Reconstruct			
		0.00	3.00	5.00	7.00	512	512-R	1024	1024-R
BPD ↓	-0.17	-0.04	-0.05	-0.07	-0.08	0.02	-0.01	0.07	-0.01
FID ↓	0.00	300.98	249.67	227.14	229.08	406.39	406.26	478.91	480.34
CLIP ↑	29.50	23.16	26.70	27.68	28.16	22.73	22.75	22.72	22.71

Table 4.5 shows that samples produced with higher CFG scale intuitively receive higher likelihood. To measure how the generative process is intertwined with likelihood, the forward process and then the backward process are applied to an image, producing an image similar to the original but with features more consistent with synthetic images. This operation is termed *reconstruction*.[4] Reconstruction increases likelihood without affecting other metrics, revealing inconsistency (Table 4.5).

Lastly, this chapter investigates visual-language reasoning, an important ability for text-to-image (TTI) models. The aim is to determine whether DM likelihood can correctly match images to captions from a list of compositionally confounding captions. Although diffusion classifiers show that this is possible with proxy-likelihoods, it remains to be determined whether this also holds for exact likelihood. The ability to discern objects is evaluated on CLEVR. For each image, all confounding prompts are presented e.g. the correct prompt is "a photo of a red sphere" but instead "a photo of a red cube" is provided. Accuracy is calculated based on the number of correct captions that were given the highest likelihood. Both colour and shape are confounded, but not at the same time. Other visual-language concepts are also considered. Complementary to the CLEVR experiment, PACS is used to measure the likelihood sensitivity to object versus domain changes, for example a photo of a horse.

The results are shown in Table 4.6. It can be seen that the scores are unexpectedly low. Scores from random selection and from selecting captions with the highest CLIP score are also provided. The results on CLEVR are comparable to random choice; when reconstruction is used, shape accuracy is reduced. SDXL performs better on PACS without reconstruction and guidance. This is anomalous behaviour since the rest of the PACS results are typically 0% accuracy.

[3]This method is further explained in Appendix 4.4.1

[4]An example is given in Appendix 4.4.2

Table 4.6: Prompt accuracy on CLEVR and PACS

Params		CLEVR		PACS	
Reconstruct	CFG	Colour	Shape	Class	Domain
✗	0	12	33	86	50
	3	0	33	0	0
	5	25	33	0	0
✓	0	38	33	14	0
	3	12	0	14	0
	5	12	0	0	0
Random		12	33	11	25
CLIP-score		63	67	100	100

4.5 Summary

Given that DMs are trained to maximise the VLB of likelihood, it is important to understand the nuances of this exact likelihood. This is especially true for conditional tasks where the conditional likelihood is often modelled implicitly, such as with a MR-VP SDE, or with CFG. The results show that even reasonable assumptions should be verified, as the task and conditional ODE used have a substantial impact on the nature of DM likelihood. For Grad- TTS, the MR-VP SDE likelihood models speaker characteristics and smooth spectrograms, but not features of language. Such findings are contrary to what one would expect for a probabilistic conditional process between speech and text. For SDXL, CFG does appear to model prompt-faithful image generation, but exact likelihood from SDXL cannot be used for prompt classification. This is inconsistent with the concept of Diffusion Classifiers that use lower bounds on likelihood for classification. Hence, these quantitative results highlight a current lack of understanding of DM likelihood. This chapter provides initial evidence that more attention should be devoted to experiments and theoretical understanding of DM likelihood, especially on conditional tasks.

4.6 Additional tables

Table 4.7: Various metrics during the generative process across depth. The image distribution quality (FID) and prompt accuracy (CLIP) improve, as does the likelihood (BPD). The best results are found at 1024 resolution with a high guidance scale.

	pipe			1.00			0.75	
CFG	height	r	BPD	FID	CLIP	BPD	FID	CLIP
0	512	✗	0.02	417.58	22.58	0.02	399.30	23.62
3	512		0.02	412.40	22.60	0.02	391.65	23.90
5	512		0.02	409.33	22.67	0.02	378.74	24.18
7	512		0.02	406.39	22.73	0.02	373.26	24.31
0	512	✓	0.01	417.80	22.58	0.00	399.60	23.61
3	512		-0.00	412.18	22.60	-0.00	392.58	23.93
5	512		-0.01	408.25	22.68	-0.01	374.78	24.19
7	512		-0.01	406.26	22.75	-0.01	373.70	24.31
0	1024	✗	0.07	478.22	22.59	0.08	418.89	21.98
3	1024		0.07	478.30	22.65	0.08	399.29	22.51
5	1024		0.07	478.65	22.70	0.08	380.89	23.04
7	1024		0.07	478.91	22.72	0.08	369.91	23.49
0	1024	✓	0.06	478.22	22.59	0.08	418.82	21.97
3	1024		0.05	479.91	22.65	0.03	402.99	22.49
5	1024		0.02	480.88	22.68	0.00	384.40	23.01
7	1024		-0.01	480.34	22.71	-0.01	375.69	23.43

	pipe			0.50			0.25	
CFG	height	r	BPD	FID	CLIP	BPD	FID	CLIP
0	512	✗	-0.01	310.19	22.68	-0.01	308.96	22.23
3	512		-0.01	280.20	23.75	-0.02	279.52	23.45
5	512		-0.02	268.82	24.36	-0.02	268.72	23.95
7	512		-0.02	257.55	24.63	-0.03	257.93	24.31
0	512	✓	-0.01	309.89	22.65	-0.02	308.74	22.23
3	512		-0.02	282.78	23.87	-0.02	278.91	23.47
5	512		-0.02	269.79	24.41	-0.03	266.02	24.08
7	512		-0.02	256.47	24.63	-0.03	256.94	24.36
0	1024	✗	-0.04	300.59	23.00	-0.06	309.11	22.99
3	1024		-0.05	245.04	26.55	-0.08	251.32	26.59
5	1024		-0.07	231.44	27.72	-0.10	232.38	27.69
7	1024		-0.08	225.89	28.10	-0.12	221.40	27.97
0	1024	✓	-0.04	300.53	23.02	-0.06	308.12	22.99
3	1024		-0.06	252.33	26.55	-0.09	263.05	26.60
5	1024		-0.07	239.35	27.69	-0.11	241.89	27.66
7	1024		-0.09	231.81	27.96	-0.13	228.23	27.95

	pipe			0.00			ref	
CFG	height	r	BPD	FID	CLIP	BPD	FID	CLIP
0	512	✗	-0.01	310.24	22.55	-0.04	0.00	29.56
3	512		-0.01	278.99	23.66	-0.04	0.00	29.56
5	512		-0.02	264.01	24.35	-0.04	0.00	29.56
7	512		-0.02	257.00	24.56	-0.04	0.00	29.56
0	512	✓	-0.01	311.61	22.57	-0.04	0.00	29.56
3	512		-0.02	280.99	23.67	-0.04	0.00	29.56
5	512		-0.02	265.90	24.28	-0.04	0.00	29.56
7	512		-0.02	259.19	24.66	-0.04	0.00	29.56
0	1024	✗	-0.04	300.98	23.16	-0.17	0.00	29.56
3	1024		-0.05	249.67	26.70	-0.17	0.00	29.50
5	1024		-0.07	227.14	27.68	-0.17	0.00	29.51
7	1024		-0.08	229.08	28.16	-0.17	0.00	29.47
0	1024	✓	-0.04	301.76	23.17	-0.17	0.00	29.56
3	1024		-0.06	255.12	26.77	-0.17	0.00	29.48
5	1024		-0.07	235.62	27.73	-0.18	0.00	29.47
7	1024		-0.09	234.73	28.10	-0.18	0.00	29.48

Chapter 5

Deep noise suppression

The noisy-speech distribution is tightly coupled to clean speech, making enhancement a different task from unconditional speech and image synthesis. This observation motivates diffusion- and Schrödinger-bridge-based models for noise suppression. This chapter investigates whether Brownian noise is needed when interpolating between paired noisy and clean data distributions. As discussed in the background chapter, Brownian motion can construct a path to a high-variance Gaussian, but paired distributions may not require the same construction.

The variance of training samples can affect model optimisation: it may regularise training, while excessive variance may make optimisation less stable. Figures 3.1 and 3.4 illustrate the corresponding probability paths and parameter choices. In the stated flow-matching construction, both are independent of u.

The main result is that the flow-matching model outperforms the tested SB model on additive-noise suppression. An additional investigation introduces SB-SV, showing how the SB interpolation becomes linear as $k \to 1$.

A further finding is that these models perform well as direct data predictors. This questions the benefits of continuous depth, as it seems that training as a neural ODE outperforms a direct regressor, but once trained it can be treated as a direct regressor. Such behaviour connects the properties of continuous-depth training with data augmentation and curriculum learning.

The rest of this chapter explains metrics and the Voicebank-DEMAND dataset. Then specifics of the models are outlined. The results are displayed after with discussion. This chapter ends with concluding remarks.

5.1 Metrics

Speech quality is evaluated with intrusive and non-intrusive metrics. The intrusive speech-enhancement metrics are PESQ (Rix *et al.*, 2001), ESTOI (Jensen and Taal, 2016), and scale-invariant signal-to-distortion ratio (SI-SDR) (Le Roux *et al.*, 2019), measured in dB. The evaluation also includes non-intrusive metrics that estimate quality from the enhanced speech

alone: DNSMOS (Reddy *et al.*, 2021),[1] a neural predictor trained on human ratings (mean opinion score (MOS)), and WhiSQA, a non-intrusive MOS predictor reported to correlate with human judgments (Close *et al.*, 2024, 2025).[2] Higher scores indicate better quality for all metrics used here.

5.2 Data

All models are trained and tested on the VB-DMD dataset (Valentini-Botinhao *et al.*, 2016), a common benchmark for speech enhancement (SE) containing clean speech recordings from 28 speakers with added background noise, e.g. café, traffic. Speakers p226 and p287 are used for validation. Non-intrusive evaluation of the clean speech yields 3.53 DNSMOS and 4.53 WhiSQA.

5.3 Model

Here, path velocity denotes the derivative of the conditional path mean with respect to depth u. Static variance means variance that is independent of u. Data prediction (DP) and flow matching (FM) denote training objectives, whereas DDP denotes an inference scheme. Following Jukić *et al.* (2024), all models are trained until validation SI-SDR converges, after which the checkpoint with the best validation PESQ is selected. Unless otherwise stated, ordinary differential equation (ODE) samplers are run for 50 steps, with batch size 8 and the same STFT settings as Richter *et al.* (2025). The neural estimator F_θ employs the NCSN++ architecture (Song *et al.*, 2021b) using the same parameterisation described in Richter *et al.* (2023). All experiments use the time-domain auxiliary loss (Jukić *et al.*, 2024).

5.4 Results

The results are displayed in Table 5.1. The tested SB-SV and ICFM settings improve several metrics over the SB-VE baseline, but the table does not show improvement on every metric. Using SB-SV with $k = 0.99$ lowers PESQ and WhiSQA relative to $k = 2.6$. ICFM with depth-independent path velocity and variance improves several reported scores. For ODE inference, FM loss gives a slightly higher SI-SDR and DNSMOS than DP, while DP gives higher PESQ; for DDP, DP has the highest PESQ and FM has the highest SI-SDR. These results do not establish a single objective as superior on all metrics. Figure 5.1 compares the supplied ODE-step results. The $N = 1$ points represent one-step ODE sampling; DDP results are reported separately in Table 5.1.

Across the reported DDP and 50-step ODE results, most metrics are comparable, and some DDP scores are higher. The proposed ICFM with DP has the highest PESQ, while ICFM with FM has the highest SI-SDR. These results suggest that models trained for ODE inference can also

[1] https://github.com/microsoft/DNS-Challenge/tree/master/DNSMOS
[2] https://github.com/leto19/WhiSQA

Path	Loss	Inference	k	c	PESQ	ESTOI	SI-SDR	DNSMOS	WhiSQA
Noisy	-	-	-	-	1.97	0.79	8.4	3.05	3.11
SB-VE	DP	ODE	2.6	0.4	2.92	0.87	19.3	3.56	4.46
SB-VE	DP	ODE	0.99	0.375	2.92	0.88	19.5	3.56	4.47
SB-SV	DP	ODE	2.6	0.15	2.98	0.88	19.4	3.58	**4.51**
SB-SV	DP	ODE	0.99	0.1	2.86	0.88	19.5	3.58	4.47
ICFM	DP	ODE	-	0.1	2.98	0.88	20.1	3.59	4.49
ICFM	FM	ODE	-	0.1	2.91	0.88	20.3	**3.60**	4.50
SB-VE	DP	DDP	2.6	0.4	2.92	0.87	19.4	3.55	4.45
SB-SV	DP	DDP	2.6	0.15	2.98	0.88	19.9	3.58	4.50
SB-SV	DP	DDP	0.99	0.1	2.99	0.88	20.0	3.57	4.50
ICFM	DP	DDP	-	0.1	**3.05**	0.88	20.2	3.58	**4.51**
ICFM	FM	DDP	-	0.1	3.00	0.88	**20.4**	3.59	**4.51**

Table 5.1: Mean speech quality metrics on VB-DMD for the proposed SB-SV and ICFM systems, with FM loss and DDP inference, compared with the SB-VE (Jukić *et al.*, 2024) baseline. $k = 0.99$ induces straightness and c scales variance.

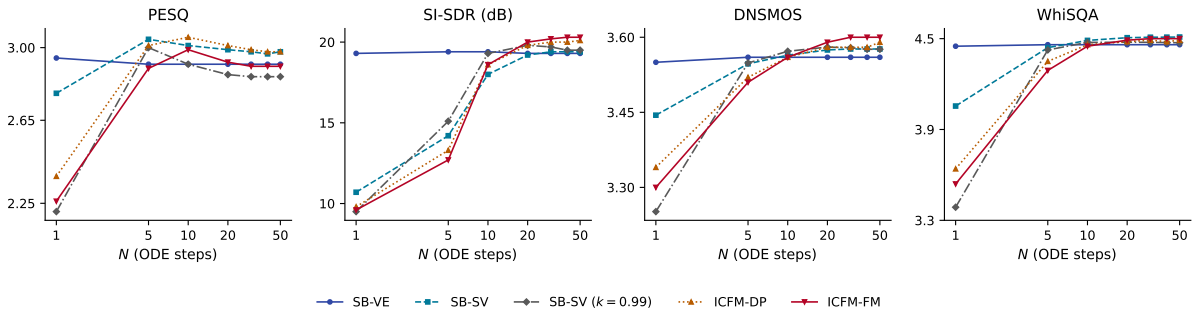


Figure 5.1: Speech-enhancement quality against the number of ODE sampling steps N on a logarithmic axis. Panels show PESQ, SI-SDR, DNSMOS, and WhiSQA. Unqualified SB-VE and SB-SV curves use $k = 2.6$; the second SB-SV curve uses $k = 0.99$. ICFM-DP and ICFM-FM denote the two training objectives. Higher scores indicate better quality.

have useful direct-prediction behaviour. The proposed explanation involving boundary variance and overfitting remains a hypothesis.

5.5 Conclusion

This chapter views the depth-independence of path velocity and variance as an analogue for straightness. This chapter assessed the impact of probability path straightness on flow-based model performance for SE. The comparison of SB-VE with SB-SV showed greater improvement with depth-independent variance than with a depth-independent path velocity. Overall, speech quality metrics improved more with ICFM, which keeps both path velocity and variance independent of depth. Using depth-independent variance degraded ODE solver performance, but this can be circumvented by directly predicting the data at inference.

Part II

Continuous time

Chapter 6

Controlled differential equations

So far, the DEs covered do not dynamically support sequence-to-sequence tasks. Let $\mathbf{x}_t \in \mathbb{R}^m$ be a control path and $\mathbf{z}_t \in \mathbb{R}^d$ the hidden state. The vector field is matrix-valued, $\mathbf{F}_\theta(\mathbf{z}, t) \in \mathbb{R}^{d \times m}$, so its product with the control increment is a hidden-state increment.

$$\mathbf{z}_t = \mathbf{z}_0 + \int_0^t \mathbf{F}_\theta(\mathbf{z}, s)\mathrm{d}\mathbf{x}_s \tag{6.1}$$

$$\mathbf{z}_t = \mathbf{z}_0 + \int_0^t \mathbf{F}_{\boldsymbol{\theta}}(\mathbf{z}, s)\frac{\mathrm{d}\mathbf{x}_s}{\mathrm{d}s}\mathrm{d}s \tag{6.2}$$

$$\mathcal{L} \coloneqq D_{\mathrm{KL}}(\mathbf{y}||\boldsymbol{g}_\phi(\mathbf{z}_T)) \tag{6.3}$$

$$\mathcal{L} = \mathbf{Y} - \mathbf{Z}_{1:T}? \tag{6.4}$$

6.1 Optimisation

Neural ODEs and CDEs can be optimised with the continuous adjoint (optimise-then-discretise) approach.

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{f}_{\boldsymbol{\theta}}(t, \mathbf{x}_t) \tag{6.5}$$

$$\mathbf{a}_{\mathbf{x}}(T) = \nabla_{\mathbf{x}_T}\mathcal{L} \quad \frac{\mathrm{d}\mathbf{a}_{\mathbf{x}}}{\mathrm{d}t} = -\left(\frac{\partial \mathbf{f}_{\boldsymbol{\theta}}}{\partial \mathbf{x}}(t, \mathbf{x}_t)\right)^\top \mathbf{a}_{\mathbf{x}}(t) \tag{6.6}$$

$$\mathbf{a}_{\boldsymbol{\theta}}(T) = 0 \quad \frac{\mathrm{d}\mathbf{a}_{\boldsymbol{\theta}}}{\mathrm{d}t} = -\left(\frac{\partial \mathbf{f}_{\boldsymbol{\theta}}}{\partial \boldsymbol{\theta}}(t, \mathbf{x}_t)\right)^\top \mathbf{a}_{\mathbf{x}}(t) \tag{6.7}$$

Here $\mathbf{a}_{\boldsymbol{\theta}}(T) = 0$ assumes that the loss has no direct dependence on $\boldsymbol{\theta}$ other than through the state. If the loss also depends on intermediate hidden states, integration restarts at each such state, adding the local loss gradient to the accumulated state adjoint $\hat{\mathbf{a}}_{\mathbf{x}}$; the parameter adjoint is carried across the restart:

$$\mathbf{a}_{\mathbf{x}}(T_i) = \hat{\mathbf{a}}_{\mathbf{x}}(T_i) + \nabla_{\mathbf{x}_{T_i}}\mathcal{L} \tag{6.8}$$

$$\mathbf{a}_{\boldsymbol{\theta}}(T_i) = \hat{\mathbf{a}}_{\boldsymbol{\theta}}(T_i) \tag{6.9}$$

$$\boldsymbol{\theta}_{j+1} \leftarrow \boldsymbol{\theta}_j - \eta \mathbf{a}_{\boldsymbol{\theta}}(0) \tag{6.10}$$

Chapter 7

Continuous-time acoustic models in speech synthesis

This chapter introduces neural CDEs to the speech synthesis task. This presents a new challenge as each sample from the continuous hidden state must be optimised, unlike more simple tasks such as speech emotion recognition where only the final state is optimised.

CDEs are a promising tool for speech synthesis as they produce a hidden state that interacts with an irregularly sampled control path. An important stage in speech synthesis is expanding the hidden state from the number of phone labels to the number of acoustic frames. The mapping between the two is monotonic and irregularly aligned, different phones have different duration due to physiological, physical and speaking style. This is the core motivation for investigating CDEs as they share this irregular mapping property.

In detail, the method lies between a text encoder and an acoustic decoder. The goal of the neural CDE is to produce time-informed features for the decoder. Such features vary with the predicted alignment of the utterance; Without a CDE, the phone features would not vary based on the time evolution of the signal, introducing a lack of variance.

The models considered in this work resolve speech-text alignment by repeating each text feature by its duration. A phone's duration can have a significant effect on how it sounds. For example, an exceptionally long vowel may become tired over time.

The integration of CDEs is evaluated thoroughly in two experimental setups. The first is a basic test on read speech, then the time-aware properties of the neural CDE is evaluated more verbosely in style-reference guided emotional speech synthesis.

The emotional speech synthesis task addressed here involves a text prompt and a speech prompt. The goal is to depict the text with the same speaker and emotion style as the speech prompt.

7.1 Data

Emotional speech dataset: Following the original work presented in the first version of StyleTTS Li *et al.* (2025), the Emotional Speech Dataset (ESD) Zhou *et al.* (2021) is used to measure emotional speech synthesis quality. The English subset of ESD comprises 350 sentences spoken by 10 speakers in 5 emotions (sad, happy, angry, neutral, surprised) totalling 10 hours. A subset of 30 minutes per speaker was selected as fine-tuning data. Random subsets of 100 samples each were held out for validation and testing, split equally across speakers and emotions. For subjective tests, the test split is reduced to 50 samples which are also split equally. The validation and test sets also contain an accompanying reference split which provides each evaluation sample with an utterance from the same speaker and emotion for style prediction. This dataset is challenging because emotional speech relies on subtle prosodic cues, such as dynamic energy and pitch control, which are perceptually salient Larrouy-Maestri *et al.* (2025) but only partially captured by TTS training losses. The use of a 5 hour subset further challenges models to capture nuanced patterns from limited data.

7.2 Training a CDE within Matcha-TTS

Matcha-TTS is an encoder-decoder architecture without style embeddings Mehta *et al.* (2024). It uses a neural ODE decoder which offers continuous computational depth. To observe how CDEs interact within TTS pipelines, a CDE is configured to use the Matcha encoder as a control path and Matcha decoder as a readout layer from hidden state to mel-spectrogram. Only the CDE is trainable, acting as a bridge between text and acoustic representation. This provides a practical test of whether a CDE can be inserted into an existing TTS pipeline without disrupting the pretrained encoder-decoder mapping.

7.3 Emotional TTS Guided by Reference Style

The core challenge addressed in this chapter involves synthesising speech in the same style as a reference sample, with attention to emotion intensity. The system uses an encoder to produce a control path for a CDE, which is then sampled and transformed to a waveform by a decoder. The non-CDE modules are initialised by training StyleTTS 2 on LibriTTS (multi-speaker audiobook narration data, neutral emotion) Li *et al.* (2023). StyleTTS 2 introduces speech language model (SLM) training, where an adversarial objective is used with WavLM features Chen *et al.* (2022), an effective yet computationally demanding process. Several memory-efficient tools are used such as mixed precision, max acoustic length of 175/800 frames, batch size of 6 without SLM adversarial training, and a batch size of 4 with SLM training. The model is trained for a total of 5 epochs, with style diffusion being trained from the second epoch and SLM from the fourth. All modules are trainable. This approach allows fine-tuning on an NVIDIA A100/H100 GPU with less than 70 GB of video memory. StyleTTS 2 checkpoints before and after fine-tuning serve as baseline models, denoted as StyleTTS 2 (LibriTTS) and StyleTTS 2 (ESD), respectively. The LSTM baseline is a separate system using a recurrent module with duration-augmented

inputs. This setup uses an ASR-based text aligner. F5-TTS is considered as a strong baseline unrelated to StyleTTS, as it has demonstrated strong zero-shot high-quality synthesis Chen *et al.* (2024). Studies with autoregressive models are left as future work. Neural CDEs can also be interpreted as continuous-time recurrent models Kidger *et al.* (2020). To examine whether the observed behaviour arises from sequential capacity rather than continuous dynamics, an experiment compares several CDE configurations with an LSTM. Both models are trained with the *duration-augmented control path* $\mathbf{X}$ provided as input.

7.4 Implementation Details

First, the initial hidden state $\mathbf{z}_0$ is predicted with an LSTM that reads the control sequence in reverse and compresses its channels. This encodes the full control context in the initial state, with the first phone $\mathbf{x}_0$ observed last: $\mathbf{z}_0 \coloneqq \mathrm{LSTM}(\mathbf{X}_{L,\dots,0})$. To reduce variation in the control variable $\mathbf{X}$, the time channel is normalised by the total phone duration for the utterance. Many CDE implementations Kidger *et al.* (2020) parameterise the vector field $\mathbf{F}_{\boldsymbol{\theta}}$ with a multilayer perceptron whose final output has dimension dC, reshaped into the matrix required by the CDE equation (Equation 6.1). With 512 control channels and hidden dimension 128, this final layer would output 65,536 values from 128 hidden units. This can be expensive. The approach proposed here uses a U-Net Ronneberger *et al.* (2015) with one downsampling stage, 256 middle channels, and 512 upsampling channels. It strides along the hidden dimension d and produces the matrix needed to evaluate the CDE (Equation 6.1). The CDE is integrated with the fixed-step fourth-order Runge–Kutta method Runge (1895); Kutta (1901). The sampled hidden sequence $\mathbf{Z}_{0,\dots,L}$ is then upsampled to 512 channels with another U-Net that processes each frame. A gated residual connection, $\mathbf{H}^{\mathrm{cde}} \coloneqq \mathbf{H}^{\mathrm{txt}} + \gamma\mathbf{Z}$, uses trainable parameter γ to integrate the CDE module with the pretrained model during fine-tuning. The Adam optimiser's β_1 is set to 0.9 for the CDE and to 0 elsewhere. The remaining setup follows the Matcha-TTS/StyleTTS 2 fine-tuning configuration Mehta *et al.* (2024); Li *et al.* (2023). The CDE-processed phone-level representations are then upsampled for the decoder. Although the reparameterisation-invariance theorem Kidger (2021) suggests that phone-level operation can provide a sufficiently expressive model, this chapter leaves open whether operating at one frame per step gives a richer temporal progression at the cost of more ODE solver steps.

7.5 Evaluation

The subjective quality of the models is evaluated with emotion intensity mean opinion score (MOS-EI) through a MUSHRA-type set up on the MTurk crowdsourcing platform. A total of 25 participants were asked to listen to a reference recording, followed by synthetic and ground truth recordings presented in a random order. The listener is only aware which sample belongs to the reference split. The reference is given to the TTS model to condition style, but not linguistic content. The participant then rates each clip on a 1-5 integer scale for how strongly the emotion is expressed in the audio e.g. "how happy is this speech?". This scheme is inspired by

Table 7.1: Effect of training a CDE as a bridge between pre-trained encoder-decoder on LJSpeech.

System	ODE Steps	MCD ↓	log-F_0 RMSE ↓
Matcha-TTS	10	5.36 ± 0.52	0.30 ± 0.08
+CDE	10	5.35 ± 0.49	0.30 ± 0.08
Matcha-TTS	5	5.30 ± 0.52	0.30 ± 0.07
+CDE	5	5.29 ± 0.48	0.29 ± 0.08

the valence-arousal model of emotion POSNER *et al.* (2005), where the label, e.g. angry, gives a hint to the target valence-arousal, and the user judges a subjective level of arousal/intensity. The mean opinion of this test reveals how strongly different systems express a target emotion, and the Spearman rank correlation of subjective scores with the reference measures reference-style tracking, while the mean absolute error (MAE) measures reference-style calibration. Overall, each sample was listened to by an average of 10 listeners. Although MUSHRA is effective at evaluating many systems, listeners pay less attention to subtle effects, motivating comparative MOS (CMOS)Li *et al.* (2023); Ju *et al.* (2024); Minixhofer *et al.* (2024). CMOS is an A/B test which displays two samples at a time, and prompts the user to rate which one is better. This evaluation uses emotion-expression quality CMOS (CMOS-EQ) with a +3/-3 integer scale to represent how much better the target emotion is expressed in system B than in system A, with negatives meaning system B is worse than system A. One of the systems will always be the candidate, and the other one of the designated baselines, displayed in random order. In both MOS tests, the listeners are prompted to focus on emotional content. Consistency was checked by duplicating 10 comparisons and swapping the A/B order without informing participants; all participants gave consistent responses for at least 5 comparisons. A question arising from these studies concerns how listeners perceive unlabelled emotional speech. This is answered by measuring emotion accuracy, which prompts the user with a simple data annotation task given both real and synthetic recordings to identify how well intended emotions are expressed by different systems, and how they are perceived. Each sample is classified by 20 participants. Objective metrics such as MCD and log-F_0 root mean squared error (RMSE) are used to measure how different design choices affect model outputs. MCD measures the difference between the mel-cepstra of generated and ground truth speech, with lower values correlating with audio quality/similarity. log-F_0 RMSE measures the error of the pitch contour of the generated signal, lower values correlate with correct intonation, emotion and prosody.

7.6 Results and Discussion

The results on LJSpeech are displayed in Table 7.1. Given that Matcha-TTS offers continuous computational depth, evaluation is conducted with both 5 and 10 ODE solver steps. The CDE produces comparable objective scores to the pretrained mapping, with small numerical differences in favour of the CDE conditions. The CDE does not disrupt the encoder–decoder mapping.

Table 7.2 shows the objective performance of short-context models (175 frames) over different step sizes (h) and stacked CDE layers. The LSTM and CDE configurations yield different

Table 7.2: Objective evaluation results of ESD for the baseline systems and CDE (short and long context).

System	MCD ↓	log-F_0 RMSE ↓
Short-context (175 frames)		
StyleTTS 2 (ESD)	5.30 ± 0.99	$\mathbf{0.40} \pm 0.13$
LSTM baseline	5.28 ± 0.98	0.41 ± 0.15
$h = 0.50$, 1 layer	$\mathbf{5.20} \pm 0.95$	$\mathbf{0.40} \pm 0.13$
$h = 0.50$, 2 layers	5.22 ± 1.01	$\mathbf{0.40} \pm 0.14$
$h = 1.00$, 1 layer	$\mathbf{5.17} \pm 0.97$	0.42 ± 0.14
$h = 1.00$, 2 layers	5.35 ± 0.96	0.42 ± 0.14
$h = 1.00$, 4 layers	5.50 ± 1.03	0.41 ± 0.15
Long-context (800 frames)		
StyleTTS 2 (ESD)	5.18 ± 0.93	0.40 ± 0.13
F5-TTS baseline	6.24 ± 1.46	0.47 ± 0.18
StyleTTS 2 (LibriTTS)	5.57 ± 0.89	0.41 ± 0.16
$h = 0.50$, 1 layer	5.20 ± 0.97	0.40 ± 0.16
$h = 0.50$, 2 layers	5.15 ± 0.97	0.40 ± 0.13
$h = 0.50$, 4 layers	5.26 ± 0.92	0.39 ± 0.14
$h = 1.00$, 1 layer	5.04 ± 0.94	0.41 ± 0.14
$h = 1.00$, 2 layers	5.07 ± 0.93	0.40 ± 0.13
$h = 1.00$, 4 layers	5.24 ± 0.93	0.40 ± 0.13

objective behaviour despite receiving the same duration-augmented inputs, indicating that the CDE is not behaving as a direct replacement for the discrete recurrent baseline. The best, second-best, and worst results are indicated by blue, orange, and red. All models benefitted from longer context. The F5-TTS baseline suffered pacing inaccuracies.

Tables 7.3, 7.4, and 7.5 show the results of the MUSHRA-style listening test. The first reports the Spearman rank correlation between a system's per-sample rating and the reference, with a macro correlation reported as an aggregate summary across emotions (computed by Fisher z-transforming the per-emotion correlations, averaging in z-space, and transforming back). The second reports the mean absolute error (MAE) between a system's intensity rating and the reference. The third reports the overall mean rating, where higher numbers indicate a higher intensity of emotion. Maximising emotion intensity does not indicate a better model, as controllable synthesis requires faithfulness to the reference emotion style, which is captured by the correlation and MAE metrics. The results show that no system is the best across all emotions, and all systems tend to under-express emotion relative to the reference, although the degree of under-expression varies across conditions. The StyleTTS 2 (ESD) baseline correlates well for surprise, is fair for neutral and happy, but negative for anger and sad. The proposed CDE models have different behaviour depending on their configuration. Using $h = 1.0$ and a single layer hinders neutral speech but improves reference-style tracking in other emotions, with poorer performance for surprise. The macro correlation of this CDE (0.53) is higher than the StyleTTS 2 (ESD) baseline (0.08). The dynamics of adjusting step size and stacking layers is non-monotonic, but it appears that stacking layers benefits correlation for $h = 0.5$, albeit a single layer has less MAE from the reference, and provides grounds for further experimentation into the dynamics of continuous-time discretisation.

Table 7.3: Per-emotion reference-style tracking: Spearman correlation with reference (↑) ($p < 0.05$ in **bold**).

System	Neu.	Ang.	Hap.	Sad	Sur.	Macro
style reference	1.00	1.00	1.00	1.00	1.00	1.00
ground truth	0.29	-0.15	0.33	0.29	0.14	0.18
F5-TTS	**0.66**	0.16	0.26	0.15	-0.02	0.26
StyleTTS 2 (LibriTTS)	0.05	0.12	0.05	0.48	0.16	0.18
StyleTTS 2 (ESD)	0.35	-0.20	0.15	-0.30	0.39	0.08
h=1.0, 1 layer	0.13	**0.64**	**0.77**	**0.70**	0.19	**0.53**
h=1.0, 2 layers	-0.16	-0.07	0.62	0.30	**0.83**	**0.38**
h=0.5, 1 layer	0.15	0.14	0.25	0.55	0.35	0.30
h=0.5, 2 layers	**0.64**	0.20	**0.64**	0.18	0.40	**0.43**

Table 7.6 shows the results of the emotion-expression quality comparison for a 1-layer CDE with $h = 1.0$. On average, listeners preferred the CDE method over the non-CDE baseline for anger and sadness, showed no preference for neutral and happy speech, and preferred the baseline for surprise. When scores were averaged per sample, the proposed and baseline systems were statistically indistinguishable under a Wilcoxon signed-rank test, indicating that the two systems produced comparable emotion-expression quality at the utterance-level. However, when averaged per listener, the results showed a significant ($p < 0.05$) overall preference for the CDE system, suggesting that listeners exhibited a consistent, albeit modest, preference for the proposed system across the evaluation set. In contrast, the CDE is significantly further than ground truth, highlighting room for improvement. Comparison with Table 7.4 suggests a partial relationship between calibration and perceived emotion-expression quality. This trend follows for surprise. Neutral-style speech contradicts this but can be considered a special case of non-emotion. A further observation is that the CDE has the lowest intensity rating for anger, yet has +0.37 CMOS-EQ. This could be a compromise where listeners prefer a less intense but more appropriately expressed realisation of anger. Finally, the results for emotion accuracy are in Table 7.7. There is a clear difference between how listeners perceived real speech versus synthetic. The StyleTTS 2 (LibriTTS) baseline overfits to neutral speech, yielding high recall. Between the other systems, the baseline is moderately favoured, with the best accuracy being the 2 layer half step CDE on sadness at 0.558. From a deployment perspective, it seems that the best strategy is to train multiple model configurations and select the optimal model for a given intended style. Together, the results indicate that CDE-based fine-tuning can improve faithfulness to the reference emotion style and may also improve perceived emotion-expression quality for some emotions, although these benefits remain dependent on emotion category and configuration.

Table 7.4: Per-emotion reference-style calibration: mean absolute rating error from reference (↓).

System	Neu.	Ang.	Hap.	Sad	Sur.	Macro
style reference	0.000	0.000	0.000	0.000	0.000	0.000
ground truth	0.227	0.499	0.292	0.447	0.268	0.347
F5-TTS	0.473	0.429	0.255	0.521	0.576	0.451
StyleTTS 2 (LibriTTS)	0.305	0.343	0.289	0.289	0.572	0.360
StyleTTS 2 (ESD)	0.281	0.482	0.288	0.396	0.418	0.373
h=1.0, 1 layer	0.298	0.426	0.214	0.225	0.543	0.341
h=1.0, 2 layers	0.311	0.405	0.262	0.331	0.555	0.373
h=0.5, 1 layer	0.246	0.419	0.355	0.157	0.403	0.316
h=0.5, 2 layers	0.157	0.538	0.356	0.310	0.517	0.376

7.7 Summary

This chapter introduced neural controlled differential equations (CDEs) as a continuous-time mechanism for duration-aware acoustic modelling in text-to-speech. The proposed method augments phone-level representations with duration-derived timing information and uses a neural acoustic vector field to produce hidden states whose values evolve with phonetic content and temporal structure, rather than using duration only for length regulation. Experiments show that CDEs provide a small benefit on neutral audiobook-style speech, but can improve reference-style tracking in emotional TTS. In particular, a single-layer CDE with $h = 1.0$ achieved the strongest macro Spearman correlation with reference emotion-intensity ratings, while CMOS-EQ results indicated comparable utterance-level emotion-expression quality and a modest listener-level preference over the fine-tuned StyleTTS 2 baseline. The results also show that step size and layer depth affect the trade-off between style tracking, calibration, and perceived emotion-expression quality, suggesting that temporal resolution is a meaningful modelling choice in CDE-based TTS. These findings motivate future investigation into alternative interpolation schemes, solver choices, frame-level control paths, and larger-scale evaluation across more diverse speakers, styles, and data resources.

Table 7.5: Absolute emotion intensity rating (MOS-EI).

System	Neu.	Ang.	Hap.	Sad	Sur.	Macro
style reference	3.59	3.54	3.49	3.43	3.62	3.53
ground truth	3.35	3.30	3.45	3.19	3.65	3.39
F5-TTS	3.09	3.51	3.31	2.96	3.10	3.20
StyleTTS 2 (LibriTTS)	3.50	3.39	3.19	3.14	3.01	3.24
StyleTTS 2 (ESD)	3.31	3.20	3.20	3.20	3.18	3.22
h=1.0, 1 layer	3.43	3.12	3.28	3.21	3.05	3.21
h=1.0, 2 layers	3.37	3.29	3.33	3.23	3.05	3.25
h=0.5, 1 layer	3.41	3.21	3.18	3.30	3.17	3.25
h=0.5, 2 layers	3.43	3.14	3.20	3.26	3.08	3.22

Table 7.6: CMOS-EQ results for CDE $h = 1$, 1 layer. Positive values indicate preference for CDE.

Baseline	Neutral	Angry	Happy	Sad	Surprise	Overall
StyleTTS 2 (ESD)	+0.03	+0.37	+0.11	+0.37	−0.28	+0.12
Ground truth	−0.50	−0.90	−1.34	−0.93	−2.01	−1.14

Table 7.7: Human emotion classification accuracy ↑.

System	Neu.	Ang.	Hap.	Sad	Sur.	Macro
style reference	0.602	0.639	0.620	0.662	0.858	0.676
ground truth	0.636	0.780	0.591	0.796	0.653	0.691
StyleTTS 2 (LibriTTS)	0.606	0.437	0.238	0.398	0.087	0.353
StyleTTS 2 (ESD)	0.537	0.399	0.304	0.479	0.209	0.386
h=1.0, 1 layer	0.478	0.492	0.245	0.469	0.175	0.372
h=0.5, 2 layers	0.422	0.318	0.294	0.558	0.255	0.369

Chapter 8

Automatic speech recognition

8.1 RNN-T

An RNN-T learns a joint probability of the next token conditioned on the text and acoustic history $P_t(y_{u+1}|\boldsymbol{y}_{1:u}, \boldsymbol{x}_{1:t})$. The probability iterates over the acoustic dimension t, and each time the joint emits a token, u increases by 1. The joint is typically implemented by representing the contexts $\boldsymbol{y}$ and $\boldsymbol{x}$ with two unimodal encoders e.g. a text predictor and an acoustic encoder

$$P_t(y_{u+1}|\boldsymbol{y}_{1:u}, \boldsymbol{x}_{1:t}) = \boldsymbol{j}(\boldsymbol{p}(\boldsymbol{y}_{1:u}), \boldsymbol{e}(\boldsymbol{x}_{1:t})). \tag{8.1}$$

There are many alignments between $\boldsymbol{y}$ and $\boldsymbol{x}$; RNN-T addresses this by using dynamic programming to optimise over all possible alignments.

8.2 Connection to coupled dynamical systems

The problem concerns describing the nature of $\boldsymbol{y}$ and $\boldsymbol{x}$.

$$y_u =? \qquad x_t =? \tag{8.2}$$

First, x_t arises from speech production, which can remain unspecified for the purposes of ASR. Second, y depends on itself and x through P_t. An equation that could satisfy this dependence is

$$y_{u+1} = f(\boldsymbol{y}_{1:u}) + g(\boldsymbol{x}_{1:t}) + c. \tag{8.3}$$

One form of coupling is specifying the rate of change of u with respect to t

$$\frac{\mathrm{d}u}{\mathrm{d}t} =? \tag{8.4}$$

This formulation allows a token to be emitted every $\Delta u = 1$, with the timing determined by the dependence of u on t.

For a streaming model, consider a hidden state $\boldsymbol{z}_t$ that emits tokens as $p(y_{u+1}|\boldsymbol{z}_t)$ when

$\Delta u = 1$. The progression of u is slower than t, and could be optimised with a neural network. If this is the case, the chain rule becomes useful.

$$\mathrm{d}\boldsymbol{z} = \boldsymbol{f}(\boldsymbol{z}, u)\frac{\mathrm{d}u}{\mathrm{d}t}\mathrm{d}t + \boldsymbol{g}(\boldsymbol{h}, t)\mathrm{d}t \tag{8.5}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} \coloneqq \zeta(\boldsymbol{z}_t, \boldsymbol{h}_t) \tag{8.6}$$

The choice of ζ gives control on how often a token is emitted. When solving this equation for $\Delta t = 1$, the Euler update rule becomes simple

$$\boldsymbol{z}_{t+1} = \boldsymbol{z}_t + \boldsymbol{f}(\boldsymbol{z}_t, u_t)\zeta(\boldsymbol{z}_t, \boldsymbol{h}_t) + \boldsymbol{g}(\boldsymbol{h}_t, t) \tag{8.7}$$

$$u_{t+1} = u_t + \zeta(\boldsymbol{z}_t, \boldsymbol{h}_t) \tag{8.8}$$

Given that a hidden state is computed for every t, CTC could be used

$$\mathcal{L} \coloneqq \mathbf{CTC}(P(\boldsymbol{y}_{1:T}|\boldsymbol{Z}_{1:T}), \boldsymbol{y}^{\mathbf{lbl}}_{1:U}) \tag{8.9}$$

A rate penalty can be added to encourage correct emissions/spikes

$$\mathcal{L} \coloneqq \int_0^U \zeta(\boldsymbol{z}_t, \boldsymbol{h}_t)\mathrm{d}t - U \tag{8.10}$$

8.3 Validation results

Model	WER	GPUs	Steps	Train time	Peak GPU memory
Streaming AED	14.89% (AED)	2	2,000	∼14 min	0.53 / 0.62 GiB
Compact MLP CTC	19.86% (frame CTC)	2	4,000	0.41 h	0.66 / 0.75 GiB
Pretrained CTC head	24.82% (frame CTC)	–	–	–	–
Joint-clock ODE	7.17% (frame CTC)	2	4,000	1.62 h	1.37 / 1.77 GiB
ODE adjoint	9.93	1	–	–	0.89 / 0.98 GiB

Table 8.1: VCTK validation results. GPU memory is peak allocated / reserved memory per GPU. The AED memory is estimated from a one-step replay and excludes DDP communication buffers. The pretrained CTC score is from a separate buffered-window diagnostic and is not directly matched to the other runs.

The encoder is frozen.

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